



Universidad Politécnica
de Madrid

**Escuela Técnica Superior de
Ingenieros Informáticos**



Master in Data Science

Master Thesis

Actual Take-Off Weight (TOW) Estimation: An Artificial Intelligence Approach

Author: Juan Andres Guevara Angel
Supervisor: Alfonso Mateos Caballero
Co-Supervisor: Faustino Tello Caballo

Madrid, July - 2025

This Master Thesis has been deposited in ETSI Informáticos de la Universidad Politécnica de Madrid.

Master Thesis

Master in Data Science

Title: Actual Take-Off Weight (TOW) Estimation: An Artificial Intelligence Approach

July - 2025

Author: Juan Andres Guevara Angel

Supervisor: Alfonso Mateos Caballero
Departamento de Inteligencia Artificial
ETSI Informáticos
Universidad Politécnica de Madrid

Co-Supervisor: Faustino Tello Caballo
Centro de Referencia I+D+i ATM (CRIDA)

Summary

This master's thesis addresses the critical challenge of accurately estimating aircraft Take-Off Weight (TOW) within the Air Traffic Management (ATM) domain. Accurate TOW data is essential for improving trajectory prediction, optimizing fuel consumption, and advancing environmental sustainability in aviation, yet it is rarely openly shared by aircraft operators due to its correlation with proprietary operational strategies. The primary objective of this research was to design, implement, and evaluate Artificial Intelligence (AI) models capable of retrospectively predicting TOW by integrating diverse data sources. These sources included operational flight data from CRIDA (Centro de Referencia de Investigación, Desarrollo e Innovación ATM), publicly available aircraft performance specifications from OpenAP, and benchmark TOW estimations derived from the EUROCONTROL Performance Review Commission (PRC) Data Challenge.

A comprehensive methodology was established, beginning with extensive data pre-processing and feature engineering. Data from Flight Plans (FPLs), flight trajectories, and aircraft specifications were merged and enriched with new features describing temporal, spatial, and dynamic aspects of each flight, with a particular focus on the climb phase. Exploratory Data Analysis revealed significant data imbalances, notably a heavy skew towards Medium (M) wake category aircraft. To mitigate this, the research was structured around three distinct case studies: (1) Medium-wake flights with available climb phase trajectory data, (2) Medium-wake flights without climb trajectory data, and (3) Heavy (H) wake category flights. This partitioning allowed for the development of specialized models tailored to the unique characteristics and data availability of each scenario. State-of-the-art Gradient Boosting Decision Tree (GBDT) algorithms: XGBoost, LightGBM, and CatBoost were utilized, alongside the H2O AutoML framework for automated model discovery and optimization using Optuna for hyperparameter tuning. MLFlow was employed for experiment tracking, ensuring reproducibility and transparency.

The results demonstrated that the developed models can predict TOW with a high degree of accuracy, successfully meeting the core objective of the thesis. For medium-wake category flights with available climb phase data, the models achieved excellent performance, with an R^2 value of approximately 0.967 and a Root Mean Squared Error (RMSE) as low as 1,333. This outcome validates the hypothesis that early flight dynamics, particularly vertical velocity during climb, are powerful predictors of aircraft mass. When climb data was absent, predictive accuracy saw a modest decline (RMSE around 1,755), but the models remained robust by leveraging other critical features like payload and cruise parameters, proving their adaptability to varying levels of data granularity. This confirms that while climb trajectory data is

highly valuable, meaningful TOW estimation is still viable without it.

The analysis of Heavy-wake (H) category aircraft presented greater challenges, primarily due to data scarcity and higher heteroscedasticity in flight operations. While the models maintained a strong correlation with actual TOW (R^2 around 0.956), the absolute prediction errors were substantially larger (RMSE around 5,233 kg), and signs of overfitting were more pronounced. This finding underscores the difficulty of modelling less frequent and more varied flight profiles and highlights a key limitation of the current dataset. Nonetheless, the models for this category are still valuable for trend analysis and strategic assessments, even if not yet suitable for precise, real-time operational use. Across all cases, aircraft-specific features such as Maximum Take-Off Weight (MTOW), Maximum Landing Weight (MLW), and Operational Empty Weight (OEW) consistently ranked as the most important predictors, reinforcing the physical and operational constraints governing aircraft weight.

In conclusion, this research demonstrates the significant potential of applying AI techniques to overcome the problem of TOW data unavailability in ATM. The resulting models provide a robust, data-driven solution that can directly support fuel consumption optimization and help achieve strategic goals outlined in ENAIRE's Flight Plan 2025 by enabling more accurate trajectory predictions.

Abstract

This Master's thesis presents an Artificial Intelligence (AI) approach for the actual Take-Off Weight (TOW) estimation of commercial aircraft, a critical parameter for accurate trajectory prediction, fuel consumption optimization, and environmental sustainability in Air Traffic Management (ATM). Given the proprietary nature and limited availability of actual TOW data, this research develops predictive models by integrating diverse data sources: operational flight data from CRIDA, public aircraft performance specifications, and benchmark TOW estimations from the EUROCONTROL Performance Review Commission (PRC) Data Challenge.

A comprehensive methodology was employed, involving extensive data preprocessing, feature engineering (incorporating temporal, spatial, and dynamic flight characteristics, with a focus on climb phase data), and an in-depth Exploratory Data Analysis (EDA). The EDA revealed significant data imbalances, particularly a predominance of Medium (M) wake category aircraft. To address this, the study was structured into three distinct case studies: (1) M-wake flights with climb phase trajectory data, (2) M-wake flights without climb trajectory data, and (3) Heavy (H) wake category flights. State-of-the-art Gradient Boosting Decision Tree (GBDT) algorithms (XGBoost, LightGBM, CatBoost) were utilized, alongside the H2O AutoML framework, with hyperparameter optimization facilitated by Optuna and experiment tracking managed by MLFlow.

The models demonstrated high predictive accuracy, particularly for M-wake flights with climb phase data, achieving an R^2 of approximately 0.967 and a Root Mean Squared Error (RMSE) of 1,333. This highlights the strong predictive power of early flight dynamics. For M-wake flights without climb trajectory data, performance remained robust (RMSE around 1,755, $R^2 > 0.95$), demonstrating adaptability to varying data granularity. H-wake category flights presented greater challenges due to data scarcity and higher variability, resulting in larger absolute errors (RMSE around 5,233) despite a respectable R^2 of 0.956. Feature importance analysis consistently identified aircraft-specific characteristics (MTOW, MLW, OEW) as primary predictors across all scenarios.

In conclusion, this research successfully demonstrates the potential of AI to provide robust, data-driven TOW estimations, directly supporting enhanced ATM efficiency and environmental objectives. While further research is needed to address data limitations for heavy aircraft and explore advanced hybrid modelling approaches, these models offer a valuable solution for operational planning and strategic assessments in aviation.

Acknowledgements

My sincere gratitude and appreciation goes to all my supervisors, who have contributed from both an academic but also professional perspective. On CRIDA's side, I would like to thank Faustino Tello for guiding and entrusting me with this project. Thank you for offering me all the support I could ever ask for and for providing me with all the necessary information, data, and expertise. On UPM's side, I would like to thank Alfonso and Antonio. Thank you for your guidance, and constant feedback, throughout the course of this MSc thesis. You have helped me gain confidence and get my ideas into shape. Most importantly, I want to thank my family. Many thanks to my parents, Ofelia and Freddy for always encouraging me and for showing me the importance of hard work and dedication.

This research was developed as part of the project of I+D+i PID2021-122209OB-C31 financed by MCIU/AEI/10.13039/501100011033.

Contents

1	Introduction	3
1.1	Objectives	6
2	Problem Statement	7
3	Literature Review	11
4	Background Information	15
4.1	Machine Learning	15
4.2	Gradient Boosting Decision Tree (GBDT)	17
4.3	Extreme Gradient Boosting (XGBoost)	21
4.4	Light Gradient Boosting Machine (LightGBM)	22
4.5	Categorical Boosting (CatBoost)	23
4.6	Optuna	24
4.7	H2O AutoML	25
4.8	MLOps & MLFlow	26
5	Data	29
5.1	Flight Plans (FPLs)	29
5.2	Flight Trajectories	30
5.3	Flight Phases	31
5.4	Plane Seats Configuration	31
5.5	Aircraft Data	32
5.6	PRC Data Challenge Dataset	32
6	Methodology	35
6.1	Data Preprocessing	36
6.2	Feature Engineering	38
6.2.1	Phase-Based Feature Engineering	38
6.2.2	Flight Trajectory Feature Engineering	39
6.2.3	Climb Velocity Feature Engineering	41
6.3	Exploratory Data Analysis	42
6.4	Case Studies	48
6.5	Predictive Models	49
6.5.1	XGBoost	50
6.5.2	LightGBM	51
6.5.3	CatBoost	52
6.5.4	Evaluation Metrics	53
6.5.4.1	Coefficient of Determination	53

6.5.4.2	Mean Absolute Error (MAE)	54
6.5.4.3	Root Mean Squared Error (RMSE)	54
6.5.4.4	Mean Absolute Percentage Error (MAPE)	54
6.5.5	MIFlow Tracking	55
7	Results	57
7.1	Flights with Climb Phase Trajectory and Medium (M) Wake Category Case	57
7.1.1	Results Analysis	58
7.2	Flights without Climb Phase Trajectory and Medium (M) Wake Category Case	59
7.2.1	Results Analysis	62
7.3	Flights with Heavy (H) Wake Category Case	62
7.3.1	Results Analysis	64
7.4	Results applicability	67
8	Conclusions and Future Work	69
	References	71

List of Abbreviations

- **ACC:** Area Control Center
- **AGI:** Artificial General Intelligence
- **AI:** Artificial Intelligence
- **AIRE:** Atlantic Interoperability Initiative to Reduce Emissions
- **ALC:** Aviation Learning Center
- **ALDT:** Actual Landing Time
- **ANI:** Artificial Narrow Intelligence
- **ANN:** Artificial Neural Network
- **ANP:** Aircraft Noise and Performance
- **ANSP:** Air Navigation Service Provider
- **ASI:** Artificial Super intelligence
- **ASPIRE:** Asia and South Pacific Initiative to Reduce Emissions
- **ATC:** Air Traffic Control
- **ATM:** Air Traffic Management
- **ATOT:** Actual Take-Off Time
- **AutoML:** Automated Machine Learning
- **CAEP:** Committee on Aviation Environmental Protection
- **CatBoost:** Categorical Boosting
- **CI:** Cost index
- **CRIDA:** Centro de Referencia de Investigación, Desarrollo e Innovación ATM
- **EDA:** Exploratory Data Analysis
- **EASA:** European Union Aviation Safety Agency
- **FDR:** Flight Data Recorder
- **FIR:** Flight Information Region
- **FPL:** Flight Plan
- **GBDT:** Gradient Boosting Decision Tree
- **GBM:** Gradient Boosting Machine
- **GBR:** Gradient Boosting Regression
- **GBRT:** Gradient Boosted Regression Trees
- **GFS:** Global Forecast System
- **GPR:** Gaussian Process Regression
- **HPO:** Hyperparameter Optimization

- **IATA:** International Air Transport Association
- **ICAO:** International Civil Aviation Organization
- **INECO:** Ingeniería y Economía del Transporte
- **LightGBM:** Light Gradient Boosting Machine
- **MAE:** Mean Absolute Error
- **MAP:** Maximum a Posteriori
- **MAPE:** Mean Absolute Percentage Error
- **MFC:** Maximum Fuel Capacity
- **ML:** Machine Learning
- **MLPNN:** Multi-Layer Perceptron Neural Network
- **MLW:** Maximum Landing Weight
- **MLOps:** Machine Learning Operations
- **MTOW:** Maximum Take-Off Weight
- **MUAC:** Maastricht Upper Area Control Center
- **OEW:** Operational Empty Weight
- **OSN:** OpenSky Network
- **PBN:** Performance-Based Navigation
- **PM:** Payload Mass
- **PRC:** Performance Review Commission
- **QAR:** Quick Access Recorder
- **RMSE:** Root Mean Squared Error
- **R²:** R-squared
- **RNAV:** Area Navigation
- **RNP:** Required Navigation Performance
- **SESAR:** Europe's Single European Sky ATM Research
- **TOW:** Take-Off Weight
- **UPM:** Universidad Politecnica de Madrid
- **XGBoost:** Extreme Gradient Boosting

Chapter 1

Introduction

Air transportation has become an indispensable component of the global economy, facilitating trade, tourism, and connectivity on a remarkable scale. However, its rapid growth has also brought significant environmental concerns, primarily related to fuel consumption and associated emissions. Ideally, air travel would involve aircraft flying optimal four-dimensional trajectories that account for variables such as wind, altitude, and speed, thereby achieving the lowest possible fuel burn and flight time. In practice, real-world operational constraints, including aircraft holding and air traffic control directives, frequently force aircraft to deviate from these highly efficient paths. Such deviations inevitably lead to increased fuel consumption and a larger environmental footprint than what is theoretically achievable. Air Traffic Management (ATM), a complex and dynamic ecosystem involving a multitude of stakeholders, including airlines, airports, Air Navigation Service Providers (ANSPs), and regulatory authorities, plays a pivotal role in mitigating these environmental impacts by actively minimizing inefficiencies in flight trajectories.

The escalating global focus on the environmental repercussions of ATM is emphasised by prominent international initiatives. Programs such as the Asia and South Pacific Initiative to Reduce Emissions (ASPIRE) [1] and the Atlantic Interoperability Initiative to Reduce Emissions (AIRE) [2] exemplify this commitment. Both initiatives aim to quantify the additional fuel burn attributable to current ATM practices in oceanic airspaces while simultaneously striving to enhance interoperability among international ATM providers. Some ATM methods can also support the fuel burn and emissions reduction, as is the case with Performance-Based Navigation (PBN) [3], which leverages satellite-based navigation systems like GPS, enabling aircraft to execute more precise and efficient routes. This encompasses both Required Navigation Performance (RNP) and Area Navigation (RNAV) procedures, facilitating more direct flight paths, optimized approach procedures, and the ability to circumnavigate adverse weather conditions, all of which contribute directly to substantial fuel savings. Recognizing this transformative potential, a growing number of ANSPs are proactively integrating fuel burn and carbon dioxide reduction targets into their official environmental performance metrics. An example of this is the Committee on Aviation Environmental Protection (CAEP), which has set an ambitious target to reduce CO_2 emissions to 2019 levels by 2027 [4]. Concurrently, major ATM modernization programs, such as Europe's Single European Sky ATM Research (SESAR) [5] and the US's Next Generation Air Transportation System (NextGen) [6], although encompass-

ing broader environmental objectives including noise reduction, improved air quality, and climate change mitigation, both unequivocally identify ATM improvement as an indispensable element for achieving their overall goals.

To further minimize ATM's environmental impacts, in the present and for the future, a key strategic approach being adopted by ATM providers is the accurate prediction of the Take-Off Weight (TOW). TOW is a parameter directly influencing take-off performance, and crucially, it has a direct impact on fuel consumption. Consequently, accurate TOW data is essential for precise trajectory predictions. While most operational and regulatory organizations are actively working to improve flight planning and emissions calculations, the absence of consistently reliable TOW data significantly compromises the accuracy of such predictions. Unfortunately, despite its profound importance, TOW data is rarely openly shared by aircraft operators. This reluctance comes from its close correlation with an airline's cost index, which can inadvertently reveal proprietary operational strategies. Disclosing such sensitive data could potentially undermine an airline's competitive positioning and even lead to regulatory or contractual penalties. The development of robust and accurate TOW prediction models, therefore, represents a step towards enhancing environmental performance in air traffic management without compromising the competitive integrity of airlines.

This Master thesis was conducted as part of an internship in collaboration with CRIDA (Centro de Referencia de Investigación, Desarrollo e Innovación ATM), a non-profit economic interest grouping formed by ENAIRE, the Universidad Politécnica de Madrid (UPM), and Ingeniería y Economía del Transporte, S.A. (INECO). Established in 2008, CRIDA functions as a public-private partnership dedicated to supporting the innovation needs of ENAIRE, the Spanish air navigation service provider. Its core mission is to enhance the overall performance of the ATM system through rigorous applied research and the development of innovative technological solutions, with a resolved commitment to prioritizing safety, efficiency, and sustainability in air navigation services.

In alignment with this mission, CRIDA plays a significant role in supporting the Flight Plan 2025 [7], ENAIRE's strategic roadmap for the period 2021–2025. This comprehensive plan was meticulously developed to ensure the long-term viability and continuous modernization of the Spanish air navigation system, particularly in response to the profound structural changes and pressing environmental challenges confronting the global aviation sector. The Flight Plan 2025 explicitly outlines ENAIRE's commitment to several key strategic pillars, including ecological transition, digital transformation, airspace efficiency, and the de-carbonization of aviation services. CRIDA's involvement in this thesis is directly framed by its strategic alignment with the objectives within the Flight Plan 2025. One of the central challenges addressed in the Plan is the imperative need to improve the environmental sustainability of ATM by actively minimizing emissions and optimizing airspace operations. In this context, the accurate prediction of TOW is a critical factor, as it directly influences aircraft performance models, fuel consumption predictions, and consequently, the capacity to effectively reduce carbon emissions.

This thesis directly supports one of the environmental pillars of the Flight Plan 2025, specifically stated through the Green Sky Strategic Plan (Strategic Plan 3). This initiative is designed to foster more sustainable operations by enabling fuel-efficient

trajectories, significantly reducing airborne delays, and promoting environmentally conscious decision-making based on robust data. Additionally, this project directly responds to the plan's pronounced focus on leveraging cutting-edge digital technologies and Artificial Intelligence (AI), as highlighted in the Digital Sky Plan (Strategic Plan 2), with the ultimate goal of achieving a more flexible, scalable, and intelligent ATM system. From CRIDA's perspective, this research aligns with its objective to function as ENAIRE's primary innovation engine, as outlined in the Innovation Strategic Plan (Strategic Plan 10). By directly contributing to the optimization of flight trajectories through improved TOW prediction, this thesis would not only enhance operational performance but also would reinforce ENAIRE and CRIDA's commitment to growth strategies like the Single European Sky initiative [8], and the 2030 Sustainable Mobility Strategy [9] .

Further emphasizing the increasing importance of data-driven approaches in ATM, the European Organization for the Safety of Air Navigation, commonly known as EUROCONTROL [10], plays a crucial role. This pan-European intergovernmental organization is dedicated to supporting both civil and military aviation across Europe, with its primary mission being the coordination of Air Traffic Management procedures and systems throughout the continent. Established in 1960, EUROCONTROL currently encompasses 41 member states and two states with comprehensive agreements, totalling 43 participating countries [11]. Headquartered in Brussels, Belgium, EUROCONTROL also operates three additional key facilities: the Innovation Hub in Brétigny-sur-Orge, France; the Aviation Learning Center (ALC) in Luxembourg; and the Maastricht Upper Area Control Center (MUAC) in Maastricht, Netherlands. Although EUROCONTROL is not an official institution of the European Union (EU), it maintains a robust collaborative relationship with the EU and works closely with a wide spectrum of aviation stakeholders, including airlines, Air Navigation Service Providers (ANSPs), airports, and international bodies such as the International Civil Aviation Organization (ICAO) and the European Union Aviation Safety Agency (EASA).

Within EUROCONTROL, the Performance Review Commission (PRC) [12] fulfils an oversight function. Established to rigorously monitor and assess ATM performance across Europe, the PRC actively cultivates transparency and continuous improvement through the analysis and publication of key performance metrics. As an integral part of its strategy to promote innovation and data openness, the PRC launched the PRC Data Challenge, an ambitious initiative designed to engage data scientists, including those from outside the traditional aviation industry, in solving complex, real-world ATM problems using open data and advanced machine learning (ML) techniques. The 2024 edition of the PRC Data Challenge provided an extensive dataset comprising 369,013 flights operated over European airspace in 2022. These data included comprehensive flight metadata such as departure and arrival airports, specific aircraft types, and precise timestamps for take-off and arrival times. A key component of this dataset was an estimated TOW value provided for each flight. In a collaborative effort with the OpenSky Network (OSN) [13], EUROCONTROL also released high-resolution trajectory data at one-second intervals, totaling approximately 158 GB in Parquet format.

For the specific purpose of this thesis, the TOW estimations provided by the PRC Data Challenge serve as the target variable in the development of robust ML models for TOW prediction. However, it is important to note that the OSN trajectory data was not utilized in this research, as it includes features that are not available in

the operational datasets typically provided by CRIDA. Since the ultimate goal of this thesis is to develop and validate models that are readily deployable within CRIDA's existing operational context, only data sources and features that align with CRIDA's internal systems were employed. These specific data sources, along with the detailed rationale behind their selection, are further elaborated in Chapter 6. In this way, the PRC Data Challenge not only powerfully illustrates the increasing importance of data science in enhancing ATM performance but also provides a highly valuable benchmark for validating and refining TOW prediction approaches. The insights garnered from this challenge directly support the development of this thesis by offering a reliable reference dataset that can be applied within CRIDA's real-world operational environment.

1.1 Objectives

The main objective of this thesis is to design, implement, and evaluate models for retrospectively predicting aircraft TOW integrating CRIDA's operational data, publicly available aircraft performance characteristics, and benchmark TOW estimations derived from the EUROCONTROL Performance Review Commission (PRC) Data Challenge.

To achieve this goal, the following specific objectives are defined:

1. Review current literature and operational practices related to TOW estimation in ATM.
2. Review the case study and define a suitable development methodology.
3. Perform an in-depth analysis of the available data, including how predictions will be generated.
4. Select appropriate ML models and establish criteria for their comparison and evaluation.
5. Assess the applicability of the final models in an operational environment and provide recommendations for future work.

Due to data availability from CRIDA, this thesis is solely dedicated to estimating TOW for commercial flights operating within the Spanish Flight Information Region (FIR). No other flights will be included, even if their TOW data is publicly available. The Spanish FIR is geographically split into two regions: Peninsula and Canarias. These are largely separated by the airspace managed by the Lisboa and Casablanca Area Control Centers (ACCs). This airspace boundary defines the segments of flight trajectories that CRIDA has access to, as flight data is exclusively tracked within this FIR. The modeling approach was designed to be compatible with CRIDA's internal systems for potential future deployment, and the results are expected to support environmentally driven innovation in ATM.

Chapter 2

Problem Statement

In the domain of civil aviation, a Flight Plan (FPL) [14] is a standardized document, prepared in accordance with a set of Air Traffic Control (ATC) instructions, used by the pilot for navigation and safe aircraft operation. It provides essential information about the intended flight, including route, cruising level, estimated departure and arrival times, aircraft type, and equipment capabilities. FPLs are foundational to ATM operations, allowing to ATC to manage traffic flow, ensure separation, and anticipate potential conflicts. For regulatory compliance, FPLs must be submitted prior to departure to an air traffic services reporting office or, during flight, transmitted to the appropriate air traffic services unit or air-ground control radio station, unless repetitive flight plans have been pre-arranged. The specific FPL data owned by CRIDA corresponds to information transmitted during flight to an air-ground control radio station, indicating that the data represents the most current and updated version of FPL available for a given flight.

Aside for the foundational importance of FPLs for ATM operations understanding, the aviation industry faces escalating pressures to significantly reduce emissions, driven by both regulatory mandates and corporate sustainability goals. Although the primary objective of most airlines remains profit maximization, reducing fuel consumption simultaneously lowers operational costs and mitigates environmental impact, aligning economic and ecological incentives. However, achieving meaningful fuel savings requires more than just efficient aircraft and routing, it demands highly accurate trajectory prediction. To comprehend the impact of TOW on aircraft performance and trajectory prediction, it is important to consider the phases of flight, which are represented as a time-altitude progression in Figure 2.1 and consist of the following segments:

- **Taxi (Out and In):** Ground movement from the gate to the runway prior to departure, and from the runway to the gate after landing.
- **Take-Off:** Acceleration along the runway until lift-off.
- **Initial Climb:** The immediate ascent phase after take-off, involving high engine thrust and steep climb angles.
- **Climb:** Continued ascent to cruising altitude, during which engine power and fuel consumption remain high.

- **Cruise:** Level flight at optimal altitude and speed, generally the most fuel-efficient phase per distance traveled.
- **Descent:** Controlled reduction in altitude, requiring less thrust but still subject to atmospheric drag and air traffic constraints.
- **Approach:** The final alignment and configuration for landing, typically involving low-altitude maneuvering.
- **Landing:** Touchdown on the runway and deceleration.

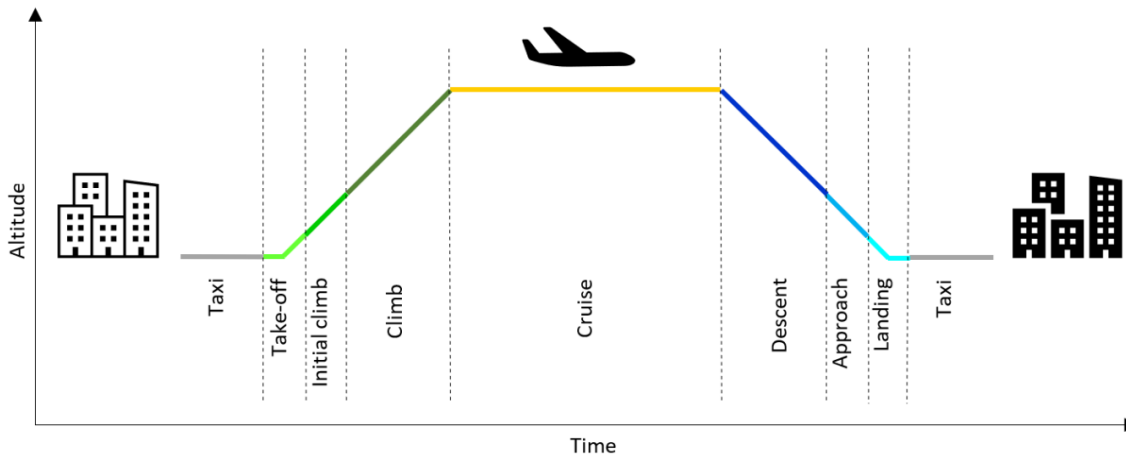


Figure 2.1: Flight phase progression over time relative to altitude [15].

These distinct flight phases highlight the dynamic nature of aircraft operations. To accurately simulate and predict aircraft movement across these phases, trajectory prediction models use point-mass dynamics, which require detailed and accurate information about the aircraft's initial state (e.g., mass, thrust setting, velocity, and position), operational intent (e.g., speed profiles, trajectory change points), and environmental conditions (e.g., temperature, wind). A significant and persistent challenge arises from the limited availability of such precise data within most ATM systems. Consequently, these models often must rely on estimations and assumptions, which inevitably introduce uncertainty and diminish predictive accuracy. Among all the parameters that influence these models, TOW stands out as one of the most critical and, unfortunately, one of the most frequently estimated rather than directly measured.

The importance of accurate TOW directly influence on an aircraft's performance during the take-off and climb phases. During these initial stages, engine thrust requirements, the necessary runway length for safe lift-off, and fuel consumption are all highly sensitive to the aircraft's weight. Operationally, underestimating TOW could lead to severe safety risks, such as insufficient take-off distance or an inability to achieve a safe climb gradient. Conversely, overestimation can result in inefficient fuel planning, carrying unnecessary fuel, or imposing tension on the airframe. From a modelling perspective, TOW is a fundamental variable that directly impacts the point-mass equations used to simulate vertical flight profiles and accurately calculate fuel consumption rates. As highlighted by EUROCONTROL and NASA, even minor errors in TOW estimations can propagate significantly through trajectory models, severely degrading their reliability, particularly during the most fuel-sensitive phases of flight [16], [17].

Problem Statement

Indeed, among the various phases of flight, take-off and climb are unequivocally the most fuel-intensive. Together, depending on the flight range, they can account for approximately 5% to 40% of the total fuel consumption [18], [19]. Although these stages are relatively short in duration compared to the cruise phase, their fuel burn per distance flown is remarkably higher. This disproportionate consumption is primarily attributable to two key factors [20]:

1. **High Engine Power Requirements:** During both take-off and the initial climb, aircraft engines must operate at near-maximum or full power to generate the immense thrust necessary to achieve the required acceleration and overcome the aircraft's substantial Take-Off Weight (TOW).
2. **Increased Aerodynamic Drag:** The higher air density prevalent at lower altitudes significantly increases aerodynamic drag. This necessitates greater thrust to maintain performance and achieve the desired climb rate, impacting not only the take-off and climb phases but also to a lesser extent, descent and landing [21].

While descent and landing are also affected by air density, they are generally less fuel-intensive than take-off and climb. This is primarily because the aircraft is considerably lighter at the end of the flight due to fuel burn, and it is actively decelerating and descending, rather than accelerating and climbing. Additionally, natural forces like aerodynamic drag and runway friction assist in deceleration, and the aircraft's potential energy is converted into kinetic energy, further reducing reliance on engine thrust. Consequently, accurate TOW is relevant for effective fuel management and impact assessment throughout the entire flight. Although the climb phase appears intuitively ideal for TOW calculation due to sustained thrust and changing altitude, all flight phases will be comprehensively included in this study to ensure that no valuable information or subtle correlations are overlooked, thus maximizing the potential for accurate predictions.

In summary, addressing the prediction of TOW is not only technically challenging but also strategically important for sustainable aviation. With all the fundamental components defined to understand the reason behind the motivation of this project, the structure of the document is given as follows: Chapter 3 reviews relevant literature, summarising key findings from previous studies and presenting the current state-of-the-art in TOW estimation. Next, Chapter 4 provides the necessary ML background information essential to understanding the development of the models. Chapter 5 describes all the datasets that will be utilized during the project development for training the models. Chapter 6 then details the methodology and process employed for the model development. This is followed by Chapter 7, which discusses the results and analyses the model's applicability. Finally, Chapter 8 concludes the thesis with final remarks and suggestions for future research directions.

Chapter 3

Literature Review

Early efforts in TOW estimation primarily relied on traditional mechanistic models or component-based approaches. For instance, Alligier, Gianazza, and Durand [22] contributed to this area by focusing on improving aircraft climb prediction through the estimation of aircraft mass. In their 2013 work, they compared two methods: an adaptive method and a least squares method. Both approaches estimated aircraft mass by fitting the modelled energy rate with the observed energy rate derived from past trajectory points. Their findings indicated that both methods provided mass estimates close to the actual mass, with the least squares method demonstrating slightly superior performance. These foundational studies were crucial in setting the stage for more sophisticated, data-driven methodologies, by demonstrating the feasibility of estimating key operational parameters from observable flight data.

In parallel, statistical modelling offered a different avenue for TOW estimation, notably in its capacity for quantifying uncertainty. Chati and Balakrishnan [23], in their MIT study, proposed a statistical approach utilizing Gaussian Process Regression (GPR). This non-parametric, probabilistic method provided not only a mean TOW estimate but also an associated confidence interval, thereby offering a crucial measure of uncertainty. Their models were developed and validated using data from Flight Data Recorders (FDRs), which are electronic devices capturing detailed flight parameters, specifically focusing on take-off ground roll segments where true TOW data was available for validation. The GPR models achieved a remarkable mean TOW error of 3% across various aircraft types, representing a nearly 50% reduction in error compared to existing Aircraft Noise and Performance (ANP) database models. While acknowledging the previous use of ML techniques for TOW modelling, they highlighted the common challenge of obtaining reliable ground truth data for accurately computing ML-based accuracy. A primary limitation of this statistical approach was its reliance on data from a single major airline, which inherently constrained the general applicability of its results across different operators or flight phases. Despite this, the study's ability to provide predictive distributions, rather than just point estimates, marked a significant advancement in understanding the inherent uncertainties in TOW modelling.

The increasing availability of diverse flight data and rapid advancements in computational power have propelled Machine Learning (ML) techniques to the vanguard of TOW estimation research. George A. Vouros et al. [24] developed a data-driven methodology that integrated mechanistic and ML models to estimate aircraft param-

eters such as Cost Index (CI) and Payload Mass (PL). They utilized DYNAMO, a trajectory prediction engine, to generate “target” Flight Plans (FPLs), which served as robust training data for their ML models. The study rigorously evaluated several ML algorithms, including lasso regression, neural networks, support vector regression, kernel ridge regression, and gradient boosting trees. Their findings indicated that gradient boosting trees offered the best balance for estimating both PL and CI, demonstrating the power of ensemble methods for complex regression tasks. Although this study did not directly estimate TOW, its comprehensive comparison of ML methods provided invaluable insights for similar estimation problems concerning other hidden aircraft parameters.

Further extending the application of ML, Richard Alligier and David Gianazza [25] specifically targeted improving aircraft climb prediction by accurately estimating TOW and speed profiles. They employed a stochastic gradient boosting tree algorithm, a choice often favoured for its strong predictive performance compared to simpler regression trees, and its insensitivity to input monotonic transformations. Their research extensively leveraged ADS-B, which refers to information automatically broadcast by aircraft using satellite navigation. The data was from The OpenSky Network, a publicly available data source, alongside weather data from the Global Forecast System (GFS). They explored two distinct approaches: predicting TOW and speed profile just before take-off, and a real-time estimation approach with a 10-minute prediction horizon using in-flight data. In their paper, Alligier and Gianazza introduced an innovative approach that conceptualized aircraft mass as the response variable of a prediction model learned from example trajectories. To avoid the challenge of unavailable actual mass data, they proposed adjusting a modelled mass for each trajectory to ensure the modelled power closely aligned with the observed energy rates. Their ML approach, encompassing Neural Networks, Gradient Boosting Machines (GBM), and other regression techniques, demonstrated significant improvements, notably reducing the Root Mean Square Error (RMSE) on predicted altitude by up to 58% compared to the BADA baseline for each aircraft type within the study. This approach of “reverse engineering” trajectory data to infer unobserved parameters is particularly relevant for the current Master’s thesis, offering a practical strategy to address the lack of direct TOW measurements.

Another notable contribution came from Xinyu He et al. [26], who proposed a hybrid approach that harmonized model-based and data-driven methods for initial-climb aircraft mass estimation. Leveraging data from Quick Access Recorders (QARs), which are airborne flight data recorders designed to provide airlines with rapid access to raw flight information, they first employed a physical model to select relevant parameters and mathematically define aircraft mass as an implicit function. Subsequently, a Multi-Layer Perceptron Neural Network (MLPNN), an ML technique known for its ability to approximate complex functions, was utilized to approximate this function. A key advantage of this innovative hybrid method is its independence from any aircraft-specific parameters, suggesting a potential for universal applicability across various aircraft types. However, its reliance on airline-specific QAR data implies that the results may not directly reflect other airline’s operational parameters, although the underlying methodology remains applicable provided access to diverse QAR datasets.

Although not exclusively focused on ML, the series of studies by Sun et al. [27], [28] contributed important methodologies and context to the task of TOW estimation.

Literature Review

These works primarily employed analytical and statistical techniques, using ADS-B data in combination with physical kinetic models. Among their methods were least-squares approximation, a mathematical approach that minimizes the sum of squared differences between observed and predicted values, and Bayesian inference, which rigorously incorporates prior knowledge through probability distributions to estimate the most probable TOW using the maximum a posteriori (MAP) principle. Their models, validated on data from 50 test flights, achieved a mean absolute error of 4.3% in mass prediction, demonstrating the effectiveness of sophisticated statistical models. While not directly utilizing ML, these studies offer valuable insights into data exploitation, assumption management, and kinetic modelling, all of which are highly relevant to the broader field of TOW estimation, particularly in understanding how to handle noisy data and incorporate physical constraints.

In summary, the current state of TOW estimation research is increasingly leaning towards data-driven methods, with ML techniques demonstrating considerable potential in enhancing prediction accuracy. Algorithms such as gradient boosting trees and neural networks, alongside hybrid model-data-driven approaches, have shown significant improvements over conventional methods. Nevertheless, substantial challenges persist, including the proprietary nature of actual TOW data, which complicates the acquisition of ground truth for ML model validation. The generalizability of models across different airlines and aircraft types remains an issue, often due to limitations in dataset diversity. However, as an advantage regarding the literature presented, CRIDA has access to comprehensive FPL data, which can significantly aid in modelling to fully account for airline-specific preferences and accurately reproduce operational intent, thereby potentially mitigating some of the noted data limitations of the literature.

Chapter 4

Background Information

An essential background knowledge necessary for the implementation of ML algorithms will be presented in this chapter. First, a general overview of AI is presented introducing all its subsets and their classes. Sections 4.1 to 4.7 then present the specific ML methods applied in the project, along with supporting tools such as Optuna, a hyperparameter optimization (HPO) framework designed to automate the search for optimal model parameters, and H2O AutoML, an automated ML framework that enables the rapid development and comparison of high-performing models with minimal user intervention. Finally, Section 4.8 introduces the concept of MLOps and describes how MLflow was employed to manage model tracking throughout the development process.

4.1 Machine Learning

ML involves developing algorithms trained on datasets to create self-learning models. These models enable computers to make predictions, decisions, or classify information without explicit programming. While the terms ML and AI are often used interchangeably, it's crucial to understand that ML is a subset of AI. AI, in its broader sense, serves as a foundational domain for several related sub-fields, representing different stages of intelligence, as depicted in Figure 4.1.

Narrow AI, also known as Artificial Narrow Intelligence (ANI), is designed for specific tasks and is currently the only achievable form of AI. Subsequent theoretical stages, Artificial General Intelligence (AGI) and Artificial Super Intelligence (ASI), have shown no progress regarding Research and Development. AGI, is a theoretical stage in which the machine would reach human intelligence, being able to replicate human-level intelligence, including problem-solving, learning, planning, and idea generation. Finally, ASI is the most advanced stage in which the machine would theoretically surpass human intelligence [29].

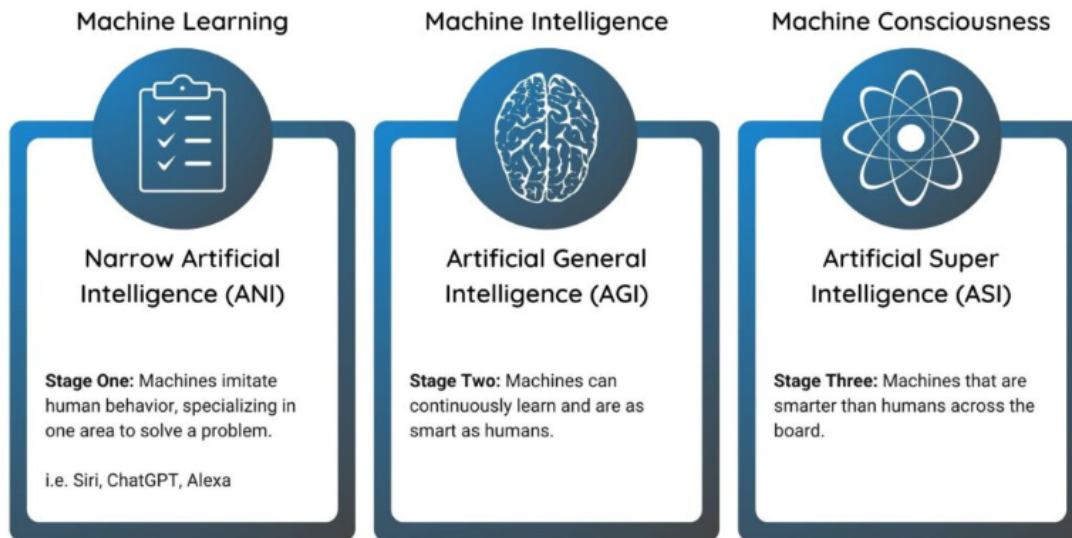


Figure 4.1: AI stages with descriptions [30].

To date, there has been no significant research and development progress towards AGI or ASI. Therefore, for the scope of this thesis, ANI will be referred to simply as AI. As previously mentioned, ML is a sub-part of AI. Figure 4.2 illustrates other areas within the AI field, where ML can be further categorized into supervised learning, unsupervised learning, and deep learning. In supervised learning, algorithms learn from labeled data, where each input (or set of inputs), known as input features, is associated with a ‘correct’ output label. On the other hand, unsupervised learning algorithms, work with unlabeled data. Their goal is not to produce a “correct” answer but to identify inherent patterns or structures within the provided dataset.

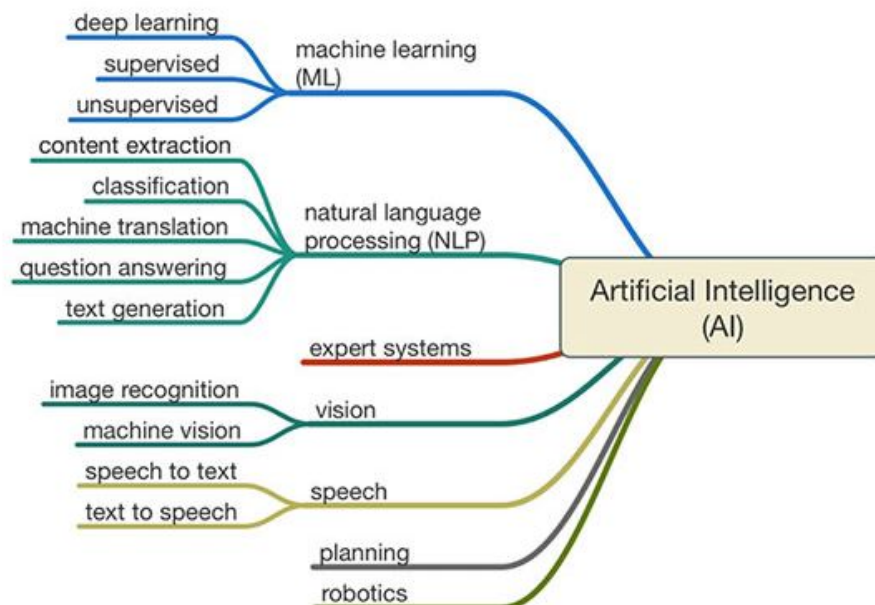


Figure 4.2: AI areas and sub-fields [31].

Background Information

Deep learning is a class of ML algorithms that utilize Artificial Neural Networks (ANNs), which are designed to mimic the structure and functionality of the human brain [32]. Key distinctions of deep learning from other ML approaches include its capacity to process larger volumes of data and its automation of feature extraction, which reduces the need for human intervention during the learning process. It's important to note that deep learning can incorporate both supervised and unsupervised algorithms. The following list briefly defines the main groups of supervised and unsupervised learning algorithms [33].

Supervised Learning:

- **Regression:** Predicts a continuous numerical value from an infinite range of possible outputs.
- **Classification:** Predicts a categorical outcome from a limited number of possible outputs.

Unsupervised learning:

- **Clustering:** Groups similar data points together from unlabeled datasets.
- **Anomaly Detection:** Identifies unusual data points, useful in applications like fraud detection.
- **Dimensionality Reduction:** Compresses large datasets into smaller ones while minimizing information loss.

The thesis will be oriented on GBDTs, which fall under the supervised learning class. The specific reasons for selecting GBDTs for this research, along with their implementation for predicting TOW, will be discussed in detail in Chapter 6.

4.2 Gradient Boosting Decision Tree (GBDT)

GBDTs are a powerful ML algorithm widely used for both classification and regression problems [34]. While GBDT is a general term, Gradient Boosting Regression (GBR) or Gradient Boosted Regression Trees (GBRTs) specifically refer to implementations designed for regression tasks, which is pertinent to this project. The core principle of GBDTs involves sequentially combining a series of 'weak' prediction models, or learners, typically decision trees, to progressively build a more accurate overall prediction of the desired output label(s). As depicted in Figure 4.3, each successive decision tree is trained to correct the errors made by the aggregate of the preceding trees [35].

An individual decision tree, as illustrated in Figure 4.4, functions similarly to a flowchart.

It is composed of decision nodes, leaf nodes, and branches. Trees are read from top to bottom, beginning with the initial decision node at the top, known as the root node. This root node, and all subsequent decision nodes, recursively partition the input features based on their values. Each branch proceeding from a decision node represents a specific decision, while leaf nodes represent the associated outcome or final prediction for that path.

The working principle of GBDTs involves five main steps, iterated sequentially:

4.2. Gradient Boosting Decision Tree (GBDT)

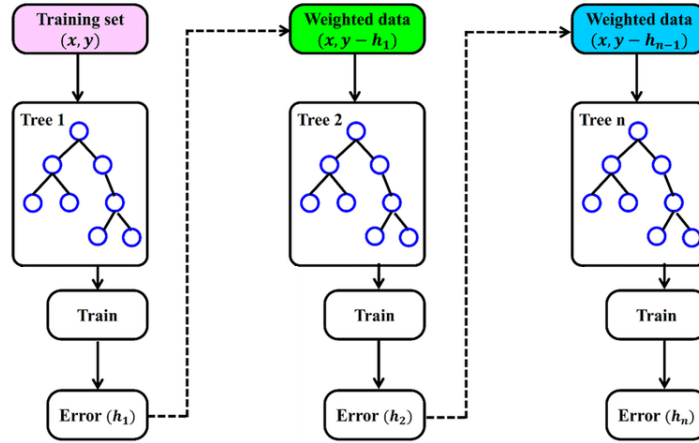


Figure 4.3: GBDT working principle [36].

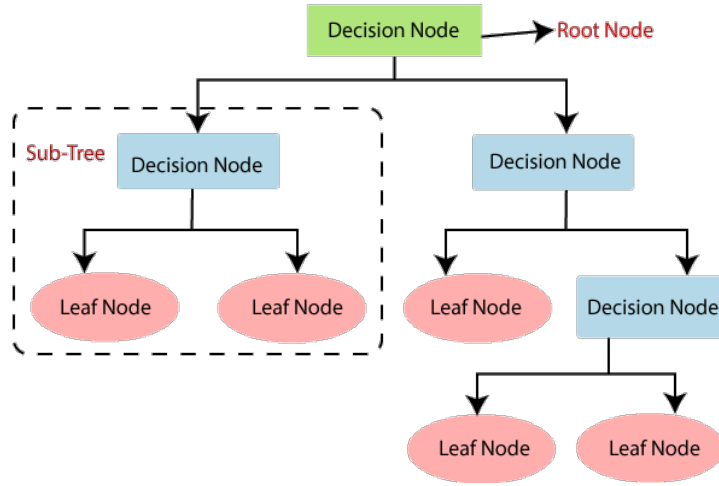


Figure 4.4: Overview of components of a decision tree [34].

1. **Initial Prediction:** The process begins by calculating the arithmetic mean of the target values x_i in the training dataset, in this case the TOWs. This initial prediction is summarized in Equation 4.1, where n represents the number of samples in the dataset.

$$\text{mean} = \frac{\sum_{i=1}^n x_i}{n} \quad (4.1)$$

2. **Residual Calculation:** For each data sample, the residual is computed using Equation 4.2. For the very first decision tree, this residual is the difference between the actual target value and the initial mean calculated in the previous step.

$$\text{residual} = \text{actual value} - \text{predicted value} \quad (4.2)$$

3. **First Decision Tree Population:** The computed residuals are used to train the first decision tree. If a leaf node contains multiple residuals, their mean is

Background Information

calculated, reducing the leaf to a single predicted residual value. It is crucial to note that the goal of this decision tree, and all subsequent ones, is to predict the residuals of the previous predictions, not the target variable (TOW) itself.

4. **Updated Prediction:** An updated residual value is assigned to each data sample based on the predictions from the newly populated decision tree. Using these updated residuals, a new predicted value for the output label (TOW) is calculated for each sample i . This is performed in step four of the algorithm using Equation 4.3, where m is the iteration number and α is the learning rate.

$$\begin{aligned} (\text{predicted value})_{m+1} = & (\text{predicted value})_m + \\ & \alpha \cdot (\text{residual predicted by decision tree})_m \end{aligned} \quad (4.3)$$

The learning rate is a critical hyperparameter introduced to prevent overfitting. As demonstrated by previous gradient boosting implementations, taking incremental steps towards the solution, rather than large jumps, introduces a slight bias. This bias, however, leads to a lower overall variance, resulting in more robust and accurate predictions for data outside the training set. By applying a learning rate ($\alpha < 1$), the model is forced to iterate more times and incorporate more decision trees, thereby facilitating this incremental approach [37].

5. **New Residuals for Next Tree:** In the fifth step, new residuals are calculated again, but this time they are based on the actual target values and the updated predicted values from the previous step (Equation 4.1, now corresponding to the step four prediction). These new residuals serve as the target for training the subsequent decision tree, as described in step three. Steps three to five are then repeated for m iterations, continuing until the prediction error, defined by a loss function, reaches an acceptable level. Once the model is fully trained, the entire sequence of decision trees is used to make a final prediction of the output label(s), computed by Equation 4.4.

$$\text{prediction} = \text{average value} + \sum_{j=1}^m \alpha \cdot (\text{residual predicted by decision tree})_m \quad (4.4)$$

The types of input features that GBDT algorithms can process, and their method of handling these features, vary depending on the specific GBDT algorithm's configuration. However, in most instances, non-numerical features, such as strings and categorical variables, must be encoded. Two common encoding methods are:

- **Integer Encoding:** Assigns a unique integer value to each category. This method is suitable for variables that possess an inherent ordinal relationship.
- **One-hot Encoding:** Used for nominal (non-ordinal) categorical variables, this method converts each unique category value into a new binary (0 or 1) variable.

The ML models chosen for this thesis are **XGBoost**, **CatBoost**, and **LightGBM**. These GBDT methods are preferred choices in data science competitions by often surpassing neural network performance on tabular data [38]. Complementing this, Murça et al. [39] offer a comprehensive comparison of these algorithms concerning key attributes

4.2. Gradient Boosting Decision Tree (GBDT)

like speed, scalability, and memory efficiency, details of which are provided in Table 4.1. Based on the robust benchmark insights from [38], the evaluation within this thesis will analyze these three GBDT methods.

Background Information

Table 4.1: Comparison of CatBoost, LightGBM, and XGBoost [39].

Feature	CatBoost	LightGBM	XGBoost
Categorical Feature Handling	Best at handling categorical variables automatically.	Requires pre-processing for categorical variables.	Requires pre-processing for categorical variables.
Overfitting Robustness	Less prone to overfitting due to symmetrical trees.	Prone to overfitting if not careful with parameters.	Regularization helps prevent overfitting effectively.
Speed	Slower, especially on large datasets.	Fastest due to histogram-based algorithms.	Fast but generally not as fast as LightGBM.
Scalability	Good but less optimal for extensive datasets.	Excellent scalability to large datasets and high-dimensional data.	Good scalability, especially with system optimizations for parallel processing.
Memory Usage	Moderate	Low due to efficient data handling.	High, especially with large data sets.
Ease of Use	Easier with default settings handling categorical data well.	Requires careful data preparation and parameter tuning.	Requires significant parameter tuning and understanding of boosting.
Data Size Handling	Handles small datasets well.	Not as effective with small datasets.	Effective with both large and small datasets but requires tuning.

4.3 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is an optimized implementation of the Gradient Boosting Decision Trees (GBDTs) algorithm developed initially by Chen and Guestrin [40]. XGBoost introduces several key optimizations that enhance its effectiveness with large-scale and high-dimensional data.

A notable advantage is its capacity for efficiently handling sparse data, obviating the need for imputation of missing or zero-filled values. To accommodate diverse data distributions, XGBoost integrates a weighted quantile sketch algorithm. This algorithm approximates quantiles by assigning different weights to data points, proving particularly valuable for datasets with varying densities or when specific data instances require prioritization, such as in imbalanced classification. Additionally,

4.4. Light Gradient Boosting Machine (LightGBM)

XGBoost's parallelized tree construction and distributed computing capabilities facilitate the processing of massive datasets with reduced training times. The algorithm also incorporates L1 and L2 regularization techniques, that penalizes the complexity of the model (the number of leaves and the magnitude of the leaf weights) and are crucial for reducing overfitting and improving model generalizability. The regularized objective function is defined by Equation 4.5 where l is the loss function and Ω is the regularization term. Collectively, these features position XGBoost as a leading algorithm in ML competitions and real-world applications, often surpassing other methods in both predictive accuracy and computational efficiency.

$$L(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (4.5)$$

4.4 Light Gradient Boosting Machine (LightGBM)

Light Gradient Boosting Machine (LightGBM), developed by Microsoft [41], is an optimized implementation of the gradient boosting principle, similar to XGBoost. It leverages the same tree-based gradient boosting approach of a GBDT but with significant improvements in training speed and efficiency.

A major enhancement in LightGBM is its use of histogram-based algorithms during training. In this approach, continuous feature values are discretized into a series of bins, specific to each feature. Bins are created for all features, and the split point within a bin that yields the greatest reduction in the loss function is selected. This histogram-based strategy is the primary reason for LightGBM's improved training speed compared to traditional gradient boosting regressors. Instead of creating a separate bin for each unique value of a feature, LightGBM creates a histogram of the feature values and then selects the best split based on the histogram [42]. This optimization also enables LightGBM models to be trained efficiently on GPUs and across multiple CPU cores, supporting parallel and distributed training.

Another key architectural difference distinguishing LightGBM from other gradient boosting algorithms, such as XGBoost and Scikit-Learn's Gradient Boosting Regressor, lies in its tree construction strategy. Unlike the level-wise tree growth strategy (shown in Figure 4.5a), LightGBM employs a leaf-wise tree growth strategy, illustrated in Figure 4.5b.

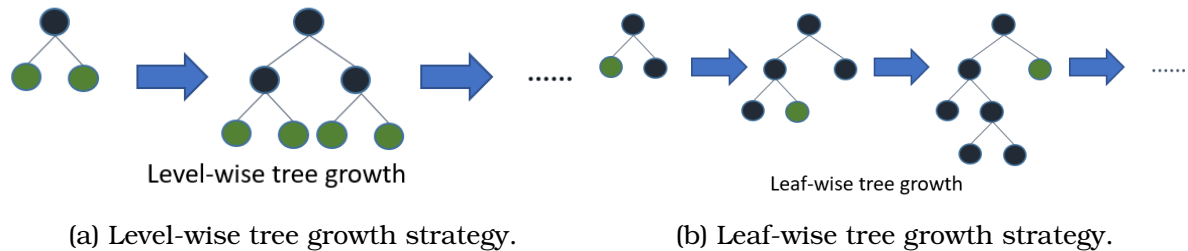


Figure 4.5: Tree Growths strategies [43].

This strategy identifies the specific leaf node that yields the minimum loss function and performs a split on that leaf only, disregarding other leaves at the same level.

Consequently, leaf-wise tree growth results in an asymmetrical tree where subsequent splitting progresses on only one side of the tree [43]. This method can further accelerate training, reduce memory usage, and enhance model accuracy. While the leaf-wise strategy often achieves lower loss function values compared to level-wise growth, it also carries a higher risk of overfitting the data, particularly with smaller datasets. Conversely, the level-wise growth strategy acts as a natural stopping mechanism for small datasets, effectively limiting tree complexity. Therefore, LightGBM's leaf-wise approach is particularly recommended for large-scale datasets where its speed and efficiency advantages are most pronounced.

4.5 Categorical Boosting (CatBoost)

CatBoost, an open-source Gradient Boosting Decision Tree (GBDT) implementation developed by Yandex [44], is specifically optimized for effectively handling categorical data in supervised ML scenarios. It introduces three primary innovations that set it apart from other GBDT algorithms:

- **Ordered Boosting:** This technique enhances the processing of data to mitigate overfitting. Unlike traditional boosting algorithms where residuals are computed based on the full dataset, Ordered Boosting calculates residuals using only a subset of the data observed before the current sample in a random permutation. This prevents a phenomenon known as “prediction shift” where the model learns from the same data it's trying to predict.
- **Ordered Target Statistics:** This is a robust technique for processing categorical variables. Instead of standard methods like one-hot encoding, statistics of the target variable are calculated for each category based on the ordered samples, thereby reducing target leakage and improving the treatment of high-cardinality categorical features.
- **Symmetric trees:** While most decision tree algorithms construct asymmetric trees where split conditions are determined independently at each node, CatBoost employs a different approach by building symmetric decision trees. This means that at any given depth level within the tree, the same split condition is applied across all branches. In fact, every leaf node at the same depth is reached by the same sequence of decisions, with different outcomes depending on the data. Figure 4.6 shows the structure of a symmetric tree, where $\leq y_1$ and $\leq y_2$ are equal split conditions.

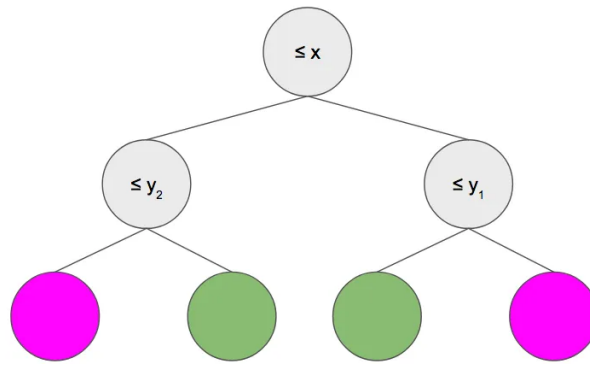


Figure 4.6: Symmetric tree [45].

These innovations enable CatBoost to manage categorical data more efficiently and robustly compared to other popular GBDT implementations, which often necessitate manual preprocessing of categorical variables into numerical formats, typically through one-hot encoding or similar techniques, which CatBoost largely automates and optimizes internally.

4.6 Optuna

Optuna is an open-source, hyperparameter optimization (HPO) framework designed for efficiently searching the vast space of potential hyperparameter configurations for ML models [46]. Unlike traditional grid search or random search methods, Optuna employs a dynamic, define-by-run API that allows for the construction of search spaces on the fly. This flexibility makes it highly adaptable to various ML models and optimization problems.

Optuna leverages advanced sampling algorithms (e.g., Tree-structured Parzen Estimator - TPE, CMA-ES) and pruning strategies to intelligently explore the hyperparameter space. Their functions are detailed below:

- Sampling algorithms propose new hyperparameter combinations for evaluation based on the performance of previously evaluated trials. This intelligent sampling allows Optuna to converge more quickly on optimal or near-optimal solutions compared to exhaustive or purely random searches.
- Pruning strategies (also known as early stopping) automatically stop unpromising trials during training. For instance, if a trial's performance is significantly worse than others at an early stage, Optuna can terminate it, thereby saving computational resources and time.

This combination of efficient sampling and intelligent pruning enables Optuna to identify high-performing hyperparameter sets with fewer computational resources and in less time than brute-force methods. Figure 4.7 illustrates the Optuna system architecture and outlines the HPO process. HPO with Optuna will be utilized for the three GBDT methods in the development of the thesis.

Background Information

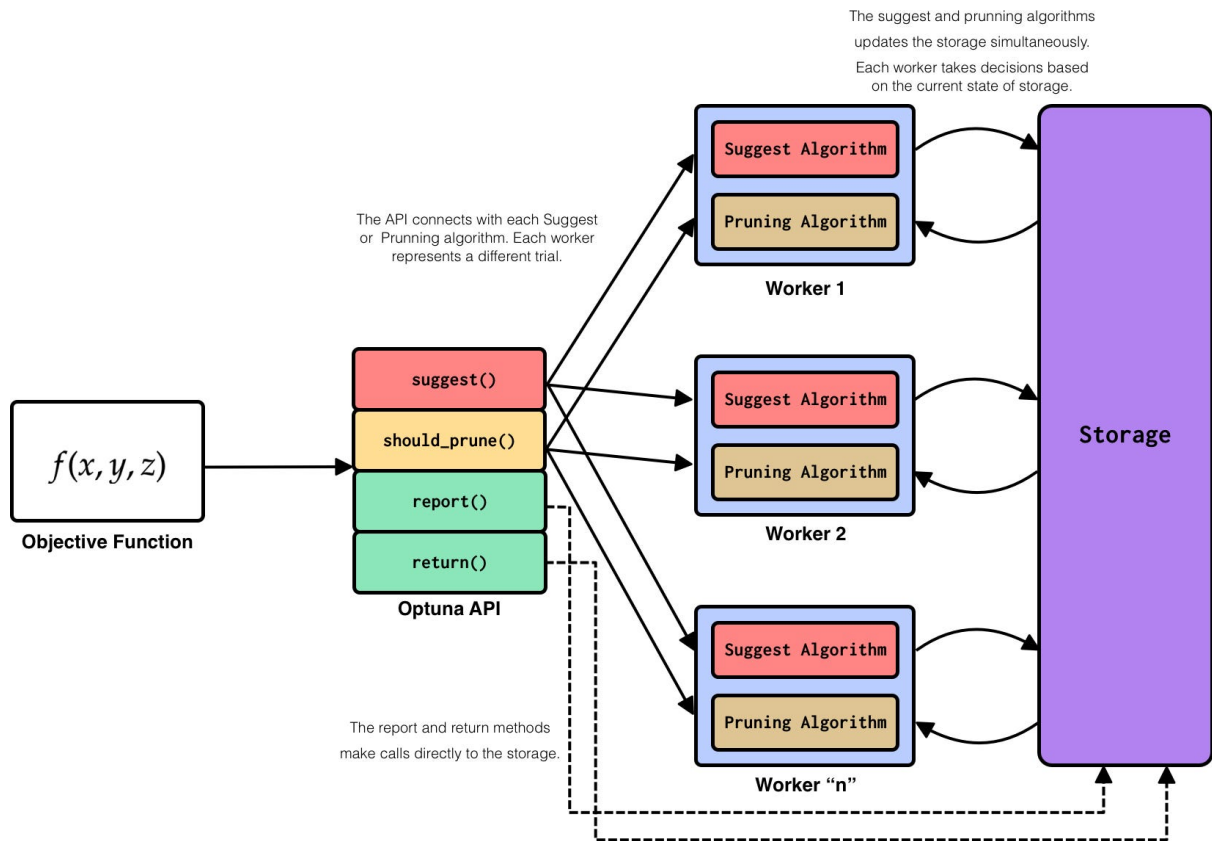


Figure 4.7: Optuna System Architecture [47].

4.7 H2O AutoML

AutoML refers to the process of automating the end-to-end steps involved in applying ML to real-world problems. Its primary goal is to make ML accessible to non-experts and to significantly accelerate the development process for experienced practitioners.

H2O AutoML is an open-source platform that incorporates these principles, streamlining the entire ML pipeline from data preprocessing to model deployment.

Developed by H2O.ai [48], it is built upon the distributed H2O-3 platform, allowing it to scale effectively for large datasets and integrate with various big data technologies. The primary objective of H2O AutoML is to automatically find the best-performing ML models for a given dataset and task (e.g., classification or regression) with minimal human intervention. The H2O AutoML process typically involves several automated steps:

1. **Data Preprocessing:** Handling missing values, encoding categorical features, and scaling numerical features.
2. **Algorithm Selection:** Automatically choosing from a wide array of algorithms, including powerful ensemble methods, as well as individual models such as: Gradient Boosting Machines (GBM), Distributed Random Forests (DRF), Generalized Linear Models (GLM), XGBoost, LightGBM, and Deep Learning (DL) models [49].
3. **Hyperparameter Tuning:** Systematically searching for optimal hyperparameters.

ters for each selected algorithm.

4. **Model Ensembling:** Building robust Stacked Ensemble models by combining predictions from the best individual models. These ensembles often achieve superior performance compared to single models.
5. **Model Leaderboard:** Providing a ranked list of all trained models based on a specified performance metric, allowing to compare and easily select the best model.

Through its automation capabilities, H2O AutoML significantly reduces time spent in hyperparameter tuning and model selection. This allows for rapid iteration to explore several model approaches.

While this thesis specifically details GBDT methods, H2O AutoML can automatically train and compare other strong candidates (including its implementations of GBM and Distributed Random Forests) and, construct powerful Stacked Ensemble models. These ensembles often achieve state-of-the-art performance by leveraging the strengths of multiple individual models, potentially offering a performance benchmark or even surpassing the individual GBDT models explored manually.

Finally, being built on the distributed H2O-3 platform, it can handle data volumes and computational demands that might challenge single-machine ML frameworks, ensuring that the methodologies explored are viable for real-world applications with extensive flight data.

4.8 MLOps & MLFlow

MLOps (Machine Learning Operations) is a set of practices that combines ML, DevOps, and Data Engineering to standardize and streamline the lifecycle of developing and deploying ML models. Its primary goal is to ensure that ML models can be reliably, efficiently, and continuously built, tested, deployed, and monitored in production environments. Beyond deployment, MLOps emphasizes the importance of reproducibility, traceability, and collaboration throughout the entire ML development process, from experimentation to production. Key aspects of MLOps include:

- **Continuous Integration (CI):** Ensures that any new feature engineering, model training script modifications, or data updates are tested and validated before being incorporated into the main development branch, thus maintaining the integrity and reproducibility of experiments.
- **Continuous Delivery (CD):** Automates the entire process of delivering a trained and validated ML model to a production or staging environment. This covers packaging the model with all its necessary dependencies (e.g., specific library versions), creating deployable artifacts (i.e., files generated during training or model execution), and automating the deployment steps to make the model available for inference. The aim is to ensure that a model, once considered ready, can be reliably and quickly released.
- **Continuous Training (CT):** Automates retraining of ML models in response to new data, concept drift, or performance degradation. Unlike traditional software, ML models degrade over time as the underlying data distribution changes. CT pipelines continuously monitor model performance and data characteristics

Background Information

in production, triggering automatic retraining when predefined conditions are met. This ensures that models remain relevant and accurate without manual intervention.

Managing the complexities of ML lifecycle and implementing these continuous practices needs a robust tool. MLflow is an open-source platform designed to manage the whole ML lifecycle, directly addressing many core MLOps challenges. Developed by Databricks [50], MLflow seamlessly integrates with widely used ML libraries such as Scikit-learn, PyTorch, Keras, and H2O. MLflow also provides a set of tools to support various stages of ML development:

- **Tracking:** This component allows users to log and store code versions, execution times, source code, hyperparameters, evaluation metrics, and artifacts. All data, except for artifacts, can be stored in databases or file systems, artifacts are stored only in file systems. MLflow supports automatic logging for most of the ML libraries. Experiments are organized into runs, each representing a single execution, and grouped into user-defined experiments.
- **Projects:** This component enables packaging of code in a reproducible and reusable manner. This module includes both an API and a command-line interface (CLI) for integrating projects into workflows. Each project consists of a Git repository or directory containing the source code, executed via defined entry points. Dependencies can be specified using virtual environments, Conda, or Docker, and are defined in a YAML configuration file named `MLproject` MLflow Documentation.
- **Models:** This component provides a standardized format for packaging ML models. Each model is stored in a directory and described by a `MLmodel` YAML file. The format supports multiple model types and automatically includes required dependencies. Models can be served directly from MLflow, and metadata must define the input and output schemas to facilitate proper integration.
- **Model Registry:** This component centralizes the management of saved models and supports collaboration throughout the model lifecycle. Models generated during experiment runs can be registered in the registry, receiving a unique name, versioning, transition states, traceability, and metadata. Registered models can exist in various lifecycle stages such as Staging, Production, or Archived. The registry can be accessed and managed via both API and a user-friendly interface, making it a vital tool for production-grade model management.

For this thesis, MLflow will play a crucial role in ensuring the reproducibility, transparency, and systematic evaluation of the developed ML models. Specifically, MLflow Tracking will be utilized to track code versions, record execution times, log hyperparameters, store evaluation metrics, and manage artifacts.

Chapter 5

Data

This chapter provides a comprehensive description of the various datasets made available for this project. All data is securely stored within CRIDA's on-premises Data Warehouse (DWH). Each section of this chapter is dedicated to a distinct dataset to which access has been granted, detailing only the information relevant to the model to be developed, even if some datasets contain a broader set of features. Furthermore, as the majority of the data CRIDA owns is mostly operational data, the reference period for processing was set from January 1, 2022, to December 31, 2022, since this is the time interval EUROCONTROL has records of TOW estimations for commercial flights operated over European airspace. The Flight Plans dataset is detailed in Section 5.1, followed by Flight Trajectories in Section 5.2, and Flight Phases in Section 5.3. Subsequently, the Plane Seats Configuration, Aircraft Data, and PRC Data Challenge Dataset are explained in Sections 5.4, 5.5, and 5.6 respectively, with only flight identifiers and TOW values retrieved from the latter, as other features are already available within DWH.

5.1 Flight Plans (FPLs)

The Flight Plans (FPLs) dataset used in this study was extracted from the "Flows" and "CFMU" data models within DWH. As highlighted in the Chapter 3, the integration of this type of data represents a notable advantage over much of the previous research on aircraft mass estimation. The dataset contains a total of 1,985,883 records and includes 20 distinct features, which are summarized in Table 5.1. These features offer comprehensive insights into both aircraft-specific characteristics and the operational intent of each flight. Aircraft-related attributes include parameters such as: `wake` and `RECATwake`, which indicate the aircraft's wake turbulence category, crucial for separation standards, and `mtow_flows`, which represents maximum certified take-off weight of the aircraft according to Flows data model. Flight-specific information captures operational intent and planned trajectories. Notable features include: `ATOT_track` and `ALDT_track`, which record the actual take-off and landing timestamps, and `cruiseSpeed` and `cruiseLevel`, representing the planned speed and cruising altitude defined in the Flight Plan.

Table 5.1: FPL data from Flows and CFMU.

Feature	Data Type	Description
flightKey	int	Unique flight identifier
callsign	string	Alphanumeric code to identify a flight
adep	string	ICAO code of the aerodrome of departure
ADEPLat	float	Latitude of the aerodrome of departure
ADEPLong	float	Longitude of the aerodrome of departure
ades	string	ICAO code of the aerodrome of destination
ADESLat	float	Latitude of the aerodrome of destination
ADESLong	float	Longitude of the aerodrome of destination
flightType	string	Type of flight (Regular, Non-Regular, GENERAL, MILITARY, Others)
wake	string	Wake turbulence category of the aircraft
RECATwake	string	RECAT wake turbulence category of the aircraft
aircraftType	string	Type of aircraft used on the flight
aircraftRegistration	string	Unique identifier assigned to an aircraft
EngineType	string	Type of Engine of the aircraft
numberOfEngines	int	Number of engines of the aircraft
aircraftRegistration	string	Unique identifier assigned to an aircraft
cruiseSpeed [kt]	int	Cruise speed defined in the FPL
cruiseLevel [FL]	int	Cruise altitude defined in the FPL
ALDT_track	timestamp	Actual landing time
arrivalRunway	string	Runway used at landing
ATOT_track	timestamp	Actual take-off time
departureRunway	string	Runway used at take-off
airlineCode	string	Identifier assigned to an airline
mtow_flows [kg]	int	Maximum take-off weight at which an aircraft is certified for take-off

5.2 Flight Trajectories

The Flight Trajectories dataset used in this study is derived from ground-based ADS-B sources covering flights within the Spanish FIR. This implies that trajectory data is available from the point an aircraft enters Spanish airspace until it exits or overflies it. ADS-B is a surveillance technology that enables aircraft to periodically broadcast their identity, position, and additional information, as appropriate. These signals can be received either by ground stations (as in this study) or by other aircraft, facilitating situational awareness, in-flight spacing, separation, and self-separation.

Due to the high frequency of ADS-B records, every five seconds, the dataset comprises a substantial number of records. These raw records were subsequently aggregated and matched to individual flight entries, as described in Section 6.2.2. The features contained in the dataset are presented in Table 5.2. Key location-based features include: `lat` and `lng` which capture the geographical coordinates of the aircraft, and `dateReference`, a timestamp indicating when each position sample was recorded.

In addition to position data, the dataset provides: `mode_c`, representing the Mode

C transponder-reported altitude in discrete flight levels, `vel_mod`, the scalar magnitude of the aircraft’s horizontal velocity vector, and `vel_z`, the vertical component of the velocity, indicating the rate of climb or descent. These features together offer a detailed view of aircraft motion dynamics within Spanish airspace and serve as a foundational component for subsequent trajectory-based analyses.

Table 5.2: Flight trajectories from Flows.

Feature	DataType	Description
adep	string	ICAO code of the aerodrome of departure
ades	string	ICAO code of the aerodrome of destination
flightKey	int	Unique flight identifier
dateReference	timestamp	Time of position
lat [°]	float	Latitude
lng [°]	float	Longitude
modo_c [fl]	int	Mode C transponder altitude
vel_mod [kt]	float	Velocity module of the aircraft
vel_z [kt]	float	Vertical velocity module of the aircraft

5.3 Flight Phases

The Flight Phases dataset, extracted from the “Flows” data model, provides temporal segmentation of each flight according to its operational phases. Specifically, it identifies three main phases of flight: “CLIMB”, “CRUISE”, and “DESCENT”, each defined by its corresponding start and end timestamps. This classification enables precise tracking of when a flight transitions between these key phases of its trajectory. The dataset contains a total of 4,723,099 records, each associated with a specific flight. The features included are summarized in Table 2.1. This dataset is crucial for understanding aircraft behavior over time, enabling phase-specific analyses.

Table 5.3: Flight Phases from Flows.

Feature	DataType	Description
flightKey	int	Unique flight identifier
phaseName	string	Flight operational phase category
dtPhaseStart	timestamp	Time of start of the phase
dtPhaseEnd	timestamp	Time of end of the phase

5.4 Plane Seats Configuration

The Plane Seats Configuration dataset, extracted from the “Flows” data model, comprises 2,163 records. This dataset is primarily characterized by its focus on aircraft seating and payload information. The core identifying feature in this dataset is `aircraftRegistration`, a string that serves as a unique identifier for each aircraft. This is preferred over `aircraftType` because the number of `seats` (an integer feature indicating the number of seats on the aircraft for a given flight) can vary even among aircraft of the same model.

Additional identifying features include `IATARegionOrigen` and `IATARegionDestino`, both strings, which categorize the `adep` and `ades`, respectively, based on International Air Transport Association (IATA) regions. These regional classifications are crucial for determining the `LoadFactor`. The `LoadFactor` is a float representing the passenger load factor, which is used to compute the `Payload`. The values for `LoadFactor` are fixed and are dependent on the IATA regional classification of both the origin and destination countries. Ultimately, the `Payload`, a float representing the carrying capacity of the aircraft (the weight of passengers, cargo, and baggage), is calculated as the product of the `seats`, the `LoadFactor`, and the average passenger weight. All the features are summarized in Table 5.4.

Table 5.4: Seats Configuration from `PlanesSeatsConfiguration_OVS`.

Feature	Data Type	Description
aircraftRegistration	string	Unique identifier assigned to an aircraft
LoadFactor	float	Passanger Load factor to compute Payload
IATARegionOrigen	string	IATA region of <code>adep</code>
IATARegionDestino	string	IATA region of <code>ades</code>
seats	int	Number of seats on the aircraft
Payload	float	Carrying capacity of the aircraft (weight of passengers, cargo, and baggage)

5.5 Aircraft Data

The Aircraft Data utilized in this study was obtained from the open-source library OpenAP [51], which provides comprehensive and transparent aircraft characteristics and engine configurations sourced from manufacturer documentation and aerospace literature. From the FPL Dataset, 25 distinct aircraft types were identified, and their corresponding characteristic features were extracted from OpenAP. The primary identifier feature in this dataset is `aircraftType`, a string that specifies the type of aircraft used on the flight. Features that are critical for TOW prediction are also included in this dataset, they are: `mtow_openap`, which represents the maximum certified take-off weight of the aircraft according to OpenAP. `mlw`, which indicates the maximum landing weight at which an aircraft is certified to land. `mfc`, that denotes the maximum fuel capacity, this is, the total weight of fuel the aircraft can carry when all tanks are full. And `oew`, which refers to the operational empty weight, which is the weight of the aircraft including crew, fluids, and essential equipment, but excluding payload and usable fuel. All the features are summarized in Table 5.5.

5.6 PRC Data Challenge Dataset

The crucial TOW values utilized in this study are derived from the PRC Data Challenge 2023 dataset [12]. This dataset encompasses estimated TOW values for several flights that operated across Europe throughout 2022. Its primary purpose in this study is to provide realistic target values, thereby enabling the development and evaluation of predictive models under conditions that closely reflect real-world flight operations. The utilized features, as summarized in Table 5.6.

Table 5.5: OpenAP library of aircraft an engine-related data.

Feature	DataType	Description
aircraftType	string	Type of aircraft used on the flight
mtow_openap [kg]	int	Maximum take-off weight at which an aircraft is certified for take-off
mlw [kg]	int	Maximum landing weight at which an aircraft is certified to land
mfc [kg]	int	The total weight of fuel that the aircraft can carry when all tanks are full
oew [kg]	int	The weight of the aircraft including crew, fluids, and essential equipment, but excluding payload and usable fuel.

Table 5.6: PRC Data Challenge Dataset.

Feature	DataType	Description
airlineCode	string	Identifier assigned to an airline
adep	string	ICAO code of the aerodrome of departure
ades	string	ICAO code of the aerodrome of destination
aircraftType	string	Type of aircraft used on the flight
ATOT_track	timestamp	Actual take-off time
tow [kg]	int	Estimated Take-Off Weight values

Chapter 6

Methodology

The methodology followed to comply with the research objective is described in this chapter. The thesis will adopt two complementary ML approaches, both of which are well-suited to structured tabular data as FPLs, Flight Trajectories, and Aircraft performance and design. The first approach relies on a set of *state-of-the-art* GB-DTs models: XGBoost, LightGBM, and CatBoost. These models have consistently demonstrated strong performance in supervised learning tasks involving structured data due to their ability to capture complex non-linear relationships, handle missing values, and provide interpretability through feature importance scores. To further enhance model performance, this approach integrates Optuna, a modern and efficient framework for HPO. Optuna utilizes a flexible and scalable optimization strategy based on Bayesian search and pruning techniques, enabling efficient exploration of large hyperparameter spaces and reducing computational overhead through early stopping mechanisms [46]. This integration allows for fine-tuning of model parameters and contributes to achieving optimal predictive performance without excessive manual intervention.

The second approach leverages the H2O AutoML framework, which offers an end-to-end automated ML pipeline tailored for tabular data. H2O AutoML automatically trains, tunes, and ranks a variety of models, of which: GBMs, DRFs, Deep Learning models, and Stacked Ensemble models will be utilized. The decision to include this framework is justified by its proven scalability, ease of use, and ability to discover high-performing models with minimal user input, which makes it particularly suitable when time and resources may be limited. AutoML frameworks like H2O have demonstrated competitive performance compared to manually tuned models, often achieving results comparable with expert-designed pipelines by efficiently combining multiple algorithms and leveraging ensemble learning techniques [38]. Furthermore, H2O's built-in support for model explainability ensures that resulting models remain interpretable and actionable in an operational ATM context.

By combining these two approaches, manual model development with rigorous hyperparameter tuning, and automated model discovery via AutoML, this methodology ensures both: granular control and efficiency, while enabling a comparative analysis of traditional and automated modelling pipelines. An overview of the overall process, including the datasets used and the modelling workflow, is presented in Figure 6.1. In this diagram, each oval box represents a dataset or key process within the pipeline.

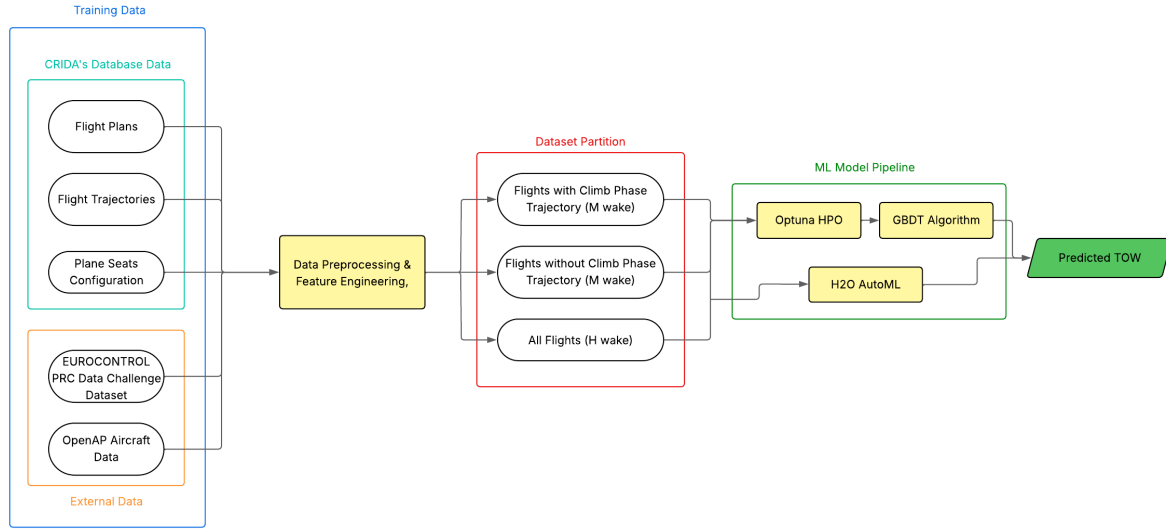


Figure 6.1: Methodology overview chart.

The training data considered in the study, including CRIDA's operational data and external datasets have already been described in Chapter 5. The steps involved in Data Preprocessing, Feature Engineering, and Exploratory Data Analysis (EDA) are covered in Sections 6.1 to 6.3. Following this, the data is partitioned into three case studies based on climb phase trajectory availability and aircraft wake category, as detailed in Section 6.4. Finally, Section 6.5 presents the ML model pipeline, including how GBDTs were optimized using Optuna for hyperparameter tuning, and applied to predict TOW, and how MLFlow logged model's crucial artifacts.

6.1 Data Preprocessing

An initial processing is a crucial step to prepare the raw flight data for analysis and model training. This involved several key transformations, including filtering irrelevant flight types, merging various datasets based on specific identifiers and timestamps, reshaping data formats, and integrating external aircraft characteristics. Each step was meticulously performed to ensure data consistency, completeness, and suitability for the modelling task. The outlined process is detailed in a step-by-step format as follows:

1. **Filtering flightType in FPLs:** Flights labelled as GENERAL and MILITARY in the flightType feature were removed from the dataset, as they correspond to non-commercial or private operations, and military flights, which fall outside the scope of this thesis.
2. **Merging FPLs with PRC Data Challenge TOW values:** To incorporate the TOW values from the PRC Data Challenge dataset into the FPLs, an asof join [52] was used. Unlike a standard left join, an asof join matches the nearest key value rather than requiring exact equality. For this to work, both datasets must be sorted by the join key. Since the FPLs and the PRC dataset do not share a common flight identifier, a combination of features, airlineCode, adep, ades, and aircraftType, was used to identify corresponding flights. Additionally,

Methodology

the timestamp feature `ATOT_track` was used as the join key, allowing matches within a 60-minute window. This means flights were matched based on how closely aligned their timestamps were, provided the identifier features matched. Figure 6.2 shows a simple example on how this kind of merge was performed. The merged dataset ended up with a size of 44,898 samples, which means that from the initial 1,985,883 FPLs dataset samples only 44,898 samples matched with an estimated TOW value.

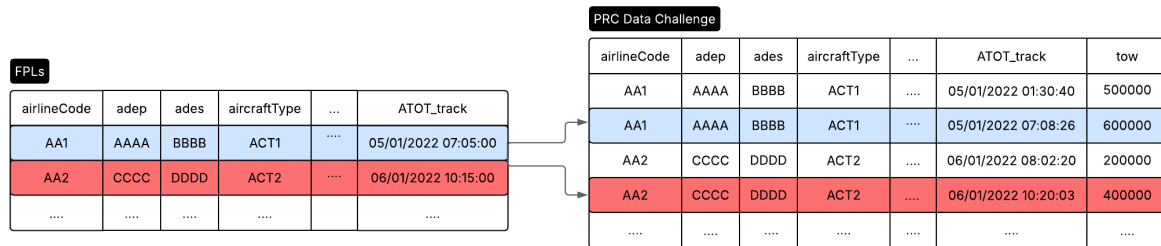


Figure 6.2: Asof join example with a 10-minute window.

- Reshaping and merging Flight Phases:** To merge the Flight Phases data with the dataset resulting from step 2, it was first necessary to reshape the Flight Phases data, which was originally in a long format. In this format, each `flightKey` appears multiple times, once for each flight phase (CLIMB, CRUISE, DESCENT). To integrate this kind of information to the previous merged dataset, a pivot transformation was utilized to reorganize the data. This pivot takes each unique `flightKey` to form new rows, while creates new columns from the `phaseName` values. For each `flightKey` and `phaseName` combination, the values from `dtPhaseStart` and `dtPhaseEnd` will populate the cells. This is illustrated in Figure 6.3. Note that the end of climb phase coincides with the start of the cruise phase, and the end of the cruise phase coincides with the start of the descending phase, since phase transitions are continuous. This also applies to `ATOT_track` and `ALDT_track`, which coincide with the start of climb phase and the end of descent phase, respectively. Once the Flight Phases were in wide format, they were merged to the previous dataset, with a regular join and `flightKey` as key the feature.
- Merging the Aircraft Data:** For last, Aircraft data was integrated. All the aircraft types of FPLs had to been passed as inputs to the OpenAP's API. The API returned almost all the feature values for each aircraft type, except for aircraft types: "BCS3", "BCS1", which are not present in OpenAP's databases. In order to fill this missing data, the values for features: `mtow_openap`, `mlw`, and `mfc`, were retrieved directly from the manufacturer manual for each aircraft type [53], [54]. For the `oew` feature values, they were retrieve for each aircraft type from *Aircraft Commerce* publication [55] about fuel burn and aircraft performance of "BCS3" and "BCS1" aircraft types.

Long Format

flightKey	phaseName	dtPhaseStart	dtPhaseEnd
11111111	CLIMB	20/03/2022 07:00:00	20/03/2022 07:12:00
11111111	CRUISE	20/03/2022 07:12:00	20/03/2022 07:37:00
11111111	DESCENT	20/03/2022 07:37:00	20/03/2022 07:57:00
22222222	CLIMB	22/04/2022 14:37:00	22/04/2022 14:51:00
22222222	CRUISE	22/04/2022 14:51:00	22/04/2022 15:06:00
....			

↓

Long Format

flightKey	dtPhaseStart_CLIMB	dtPhaseEnd_CLIMB	dtPhaseStart_CRUISE	dtPhaseEnd_CRUISE	dtPhaseStart_DESCENT	dtPhaseEnd_DESCENT
11111111	20/03/2022 07:00:00	20/03/2022 07:12:00	20/03/2022 07:12:00	20/03/2022 07:37:00	20/03/2022 07:37:00	20/03/2022 07:57:00
22222222	22/04/2022 14:37:00	22/04/2022 14:51:00	22/04/2022 14:51:00	22/04/2022 15:06:00	null	null
....						

Figure 6.3: Flight Phases dataset reshaped from long to wide format.

6.2 Feature Engineering

A feature engineering process was conducted to augment the preprocessed flight data with a diverse array of new features, primarily derived from FPL data and flight trajectory information. This enhancement converted the dataset significantly more robust, informative, and conducive to predictive modelling and comprehensive analysis.

The initial set of newly integrated features encompasses temporal attributes associated with take-off and landing timestamps, specifically the variables denoted as `ATOT_track` and `ALDT_track`. These features include the hour of the day for both take-off (`ATOT_track_hour`) and landing (`ALDT_track_hour`), in addition to the day of the year indicators (`ATOT_track_day` and `ALDT_track_day`). Such variables are anticipated to serve as critical predictors of flight behaviour patterns and performance variability. Subsequent features integrated into the dataset are phase-based and describe the spatial dynamics of flight trajectories; these features are presented in detail in the following sections.

6.2.1 Phase-Based Feature Engineering

To facilitate an accurate temporal analysis of flight records, it was essential to consider the duration of each flight phase. During the data preprocessing stage, the timestamp features `dtPhaseStart_CLIMB`, `dtPhaseEnd_CLIMB`, `dtPhaseStart_CRUISE`, `dtPhaseEnd_CRUISE`, `dtPhaseStart_DESCENT`, and `dtPhaseEnd_DESCENT` were defined, allowing for the calculation of flight phase durations. These durations were determined by calculating the number of hours between the start and end of each flight phase. However, a significant challenge arises with the start and end timestamps of flight phases, which may be either undefined or incorrectly defined when the aircraft enters or exits the Spanish FIR for phases oc-

Methodology

curing outside of it. This results in flights with international and overflight routes potentially lacking defined values for some start and end timestamps.

To address this issue, only the durations of well-defined flight phases were considered, while the missing values were imputed using the statistical median of flights within the same wake category. Consequently, the resulting variables in [hours] units included

CLIMB_duration, CRUISE_duration, DESCENT_duration, and flight_duration, with the latter defined as the total duration of the flight, calculated using ATOT_track and ALDT_track timestamps. It is important to note that missing or inconsistent start and end timestamps can be addressed by adding or subtracting the phase duration from another known timestamp, as represented in Equation 6.1.

$$\begin{aligned} \text{dtPhaseEnd_CLIMB} &= \text{dtPhaseStart_CLIMB} + \text{CLIMB_duration} \\ \text{dtPhaseStart_DESCENT} &= \text{dtPhaseEnd_DESCENT} - \text{DESCENT_duration} \end{aligned} \quad (6.1)$$

Regarding the cruise phase duration, if median value is zero, the cruise phase can also be derived by subtracting CLIMB_duration and DESCENT_duration from flight_duration. This approach ensures a comprehensive temporal representation of the flight, as illustrated in Equation 6.2.

$$\text{CRUISE_duration} = \text{flight_duration} - \text{CLIMB_duration} - \text{DESCENT_duration} \quad (6.2)$$

6.2.2 Flight Trajectory Feature Engineering

Regarding spatial dynamics, these were captured by calculating both the total flight distance, represented by the feature total_flight_distance, and phase-specific distances, indicated by the features in [NM] units: CLIMB_distance, CRUISE_distance, and

DESCENT_distance. This process initiated by querying into the Flight Trajectory Dataset, to retrieve the time-location features to identify the aircraft during flight (dateReference, lat, and lng) and the physics-based features describing the dynamics of the aircraft during the flight (modo_c, vel_mod, and vel_z).

To effectively aggregate this information, the Haversine formula [56] was employed. This formula defines the distance d as the great-circle distance between two points on a sphere based on their longitudes and latitudes, as delineated in Equation 6.3. Given that the Earth approximates a spherical shape (more accurately, an ellipsoid), the Haversine formula facilitates the calculation of the shortest distance between two points along a curved surface, thereby providing a close estimation of the distance covered by an aircraft.

$$d = 2r \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\varphi}{2} \right) + \cos \varphi_1 \cdot \cos \varphi_2 \cdot \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right) \quad (6.3)$$

Where:

- φ_1 and φ_2 are the latitude of point 1 and the latitude of point 2,
- λ_1 and λ_2 are the longitude of point 1 and the longitude of point 2,
- r is the radius of the sphere (the radius of the earth in this case),
- $\Delta\varphi = \varphi_2 - \varphi_1$, $\Delta\lambda = \lambda_2 - \lambda_1$.

To compute `total_flight_distance` [NM], the analysis utilized the coordinates of the departure aerodrome (ADEPLat, ADEPLong) and the destination aerodrome (ADESLat, ADESLong) to capture the planned route distance of flights. For phase-specific distances, challenges arose concerning the availability of trajectory data, which is limited to locations within the Spanish FIR. To address potential gaps in trajectory data for the CLIMB and DESCENT phases, the analysis filled these gaps by applying the Haversine formula between the initial or final trajectory points and the coordinates of the respective departure or destination aerodromes, as appropriate.

In the case of the CRUISE phase, the challenge was addressed by imputing the statistical median within the same `wake` category. If the wake-specific median was zero or incorrectly defined, `CRUISE_distance` was estimated by deducting `CLIMB_distance` and `DESCENT_distance` from `total_distance`, as expressed in Equation 6.4.

$$\text{CRUISE_distance} = \text{total_distance} - \text{CLIMB_distance} - \text{DESCENT_distance} \quad (6.4)$$

To aggregate the remaining features that describe the dynamics of the aircraft throughout the flight, namely `modo_c`, `vel_mod`, and `vel_z`, the statistical median and variance for each feature were computed across segments of the flight trajectory corresponding to distinct flight phases. This approach resulted in the integration of a total of 19 new features, with altitude in [fl] units and velocities in [kt] units, defined as follows:

- `CLIMB_modo_c_median`: Median pressure altitude during climb phase,
- `CLIMB_modo_c_variance`: Variance of pressure altitude during climb phase,
- `CLIMB_vel_mod_median`: Median speed during climb phase,
- `CLIMB_vel_mod_variance`: Variance of speed during climb phase,
- `CLIMB_vel_z_median`: Median vertical velocity during climb phase,
- `CLIMB_vel_z_variance`: Variance of vertical speed during climb phase,
- `CRUISE_modo_c_median`: Median pressure altitude during cruise phase,
- `CRUISE_modo_c_variance`: Variance of pressure altitude during cruise phase,
- `CRUISE_vel_mod_median`: Median speed during cruise phase,
- `CRUISE_vel_mod_variance`: Variance of speed during cruise phase,
- `CRUISE_vel_z_median`: Median vertical velocity during cruise phase,
- `CRUISE_vel_z_variance`: Variance of vertical speed during cruise phase,
- `DESCENT_modo_c_median`: Median pressure altitude during descent phase,

- DESCENT_mod0_c_variance: Variance of pressure altitude during descent phase,
- DESCENT_vel_mod_median: Median speed during descent phase,
- DESCENT_vel_mod_variance: Variance of speed during descent phase,
- DESCENT_vel_z_median: Median vertical velocity during descent phase,
- DESCENT_vel_z_variance: Variance of vertical speed during descent phase

Furthermore, the difference in altitude of `modo_c` at the beginning and end of each flight phase was incorporated as additional features: `CLIMB_delta_mod0_c`, `CRUISE_delta_mod0_c`, and `DESCENT_delta_mod0_c`. Similar to the temporal and spatial features, the availability of trajectory data is restricted to the Spanish FIR in relation to these features. Consequently, any missing values for each feature were imputed using the statistical median.

6.2.3 Climb Velocity Feature Engineering

To enhance the utilization of flight trajectory data, it has been established that features associated with the early climb phase serve as critical parameters for estimating aircraft mass and TOW [22]. Consequently, a range of early altitude intervals was defined to capture the detailed velocities that occur at the outset of the climb phase: [0-30 flt levels (flt)], [0-60 flt], [0-90 flt], [30-60 flt], [30-90 flt], and [60-90 flt]. In a manner similar to the phase-based features characterizing the dynamics of the aircraft during flight, the statistical mean and variance were computed for each of these intervals. This process resulted in a total of 24 features being integrated in [kt] units, as follows:

- CLIMB_vel_mod_mean_0-30flt: Average rate of ascent for interval [0-30flt],
- CLIMB_vel_mod_variance_0-30flt: Variance of the rate of ascent for interval [0-30flt],
- CLIMB_vel_z_mean_0-30flt: Average vertical rate of ascent for interval [0-30flt],
- CLIMB_vel_z_variance_0-30flt: Variance of vertical rate of ascent for interval [0-30flt],
- CLIMB_vel_mod_mean_0-60flt: Average rate of ascent for interval [0-60flt],
- CLIMB_vel_mod_variance_0-60flt: Variance of the rate of ascent for interval [0-60flt],
- CLIMB_vel_z_mean_0-60flt: Average vertical rate of ascent for interval [0-60flt],
- CLIMB_vel_z_variance_0-60flt: Variance of vertical rate of ascent for interval [0-60flt],
- CLIMB_vel_mod_mean_0-90flt: Average rate of ascent for interval [0-90flt],
- CLIMB_vel_mod_variance_0-90flt: Variance of the rate of ascent for interval [0-90flt],
- CLIMB_vel_z_mean_0-90flt: Average vertical rate of ascent for interval [0-90flt],
- CLIMB_vel_z_variance_0-90flt: Variance of vertical rate of ascent for interval [0-90flt],

- CLIMB_vel_mod_mean_30-60fl: Average rate of ascent for interval [30-60fl],
- CLIMB_vel_mod_variance_30-60fl: Variance of the rate of ascent for interval [30-60fl],
- CLIMB_vel_z_mean_30-60fl: Average vertical rate of ascent for interval [30-60fl],
- CLIMB_vel_z_variance_30-60fl: Variance of vertical rate of ascent for interval [30-60fl],
- CLIMB_vel_mod_mean_30-90fl: Average rate of ascent for interval [30-90fl],
- CLIMB_vel_mod_variance_30-90fl: Variance of the rate of ascent for interval [30-90fl],
- CLIMB_vel_z_mean_30-90fl: Average vertical rate of ascent for interval [30-90fl],
- CLIMB_vel_z_variance_30-90fl: Variance of vertical rate of ascent for interval [30-90fl],
- CLIMB_vel_mod_mean_60-90fl: Average rate of ascent for interval [60-90fl],
- CLIMB_vel_mod_variance_60-90fl: Variance of the rate of ascent for interval [60-90fl],
- CLIMB_vel_z_mean_60-90fl: Average vertical rate of ascent for interval [60-90fl],
- CLIMB_vel_z_variance_60-90fl: Variance of vertical rate of ascent for interval [60-90fl]

It is important to note that these features are exclusively designated for flights that exhibit complete climb trajectory segments. Given the highly variable speeds that may be encountered during the early climb phase, imputing missing values for these variables can lead to suboptimal data handling.

The decision to substitute missing values with the median is grounded in its resistance to extreme outliers, providing a more accurate representation of the central tendency in skewed data. For both phase-based and flight trajectory aggregated features, the data were aggregated utilizing the median and variance grouped by its corresponding `wake` category for each phase segment. This approach minimizes the influence of anomalous trajectories and assesses the spread or volatility of the data for aircrafts with similar performing characteristics, while also evaluating the consistency across flight phases. For the aggregated climb velocity features, the mean and variance were employed to capture both typical and extreme performance metrics, along with quantifying variability.

In summary, the feature engineering process articulated in this section aims to leverage both time-series trajectory data and static flight information, thereby creating a comprehensive array of predictive features. This process also addresses the challenges of missing values and potential inconsistencies, ultimately preparing the dataset for rigorous modelling.

6.3 Exploratory Data Analysis

An exploratory data analysis of the processed dataset was conducted to gain a comprehensive understanding of its characteristics. Figure 6.4b illustrates the total num-

ber of flight observations recorded for each month of 2022, while Figure 6.4a displays the hourly distribution of actual departure times. The monthly distribution indicates that August 2022 experienced the highest volume of flights, likely reflecting the increased demand characteristic of the European summer vacation period. Conversely, February 2022 recorded the lowest number of operations. The hourly distribution reveals a rise in traffic departing from European airports during the early morning hours, peaking at 11:00 UTC, followed by a gradual decline until the end of the day.

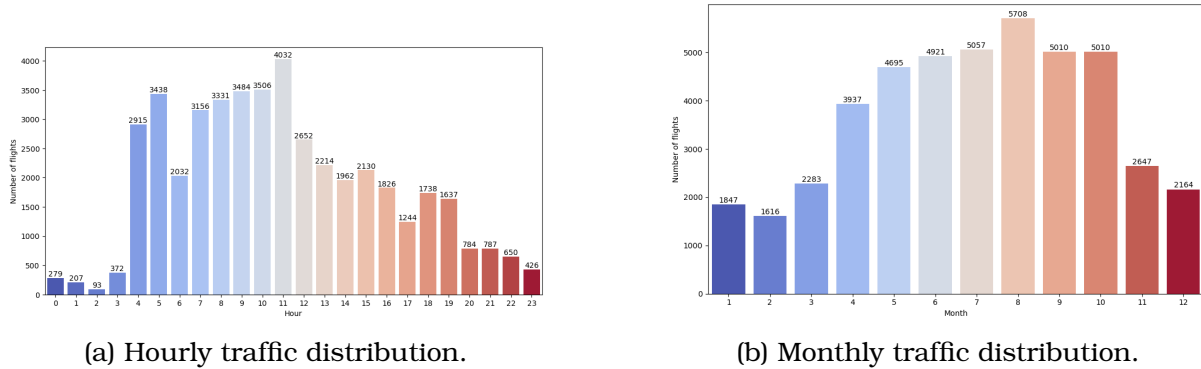


Figure 6.4: Distribution of flight traffic.

In terms of spatial coverage, the flight data encompasses observations for 931 origin-destination pairs, as depicted in Figure 6.5. A majority of the flights operate within European airspace; however, the dataset also includes international flights connecting Europe with regions such as North and West Africa, Central America, North America, and South America. Notably, a maximum of 65 flights were recorded between Alicante Elche Miguel Hernández Airport (LEAL) and Málaga Costa del Sol Airport (LEMG). Among international routes, Zurich Airport (LSZH) in Switzerland and Josep Tarradellas Barcelona–El Prat Airport (LEBL) accounted for the highest volume, with 780 flights.

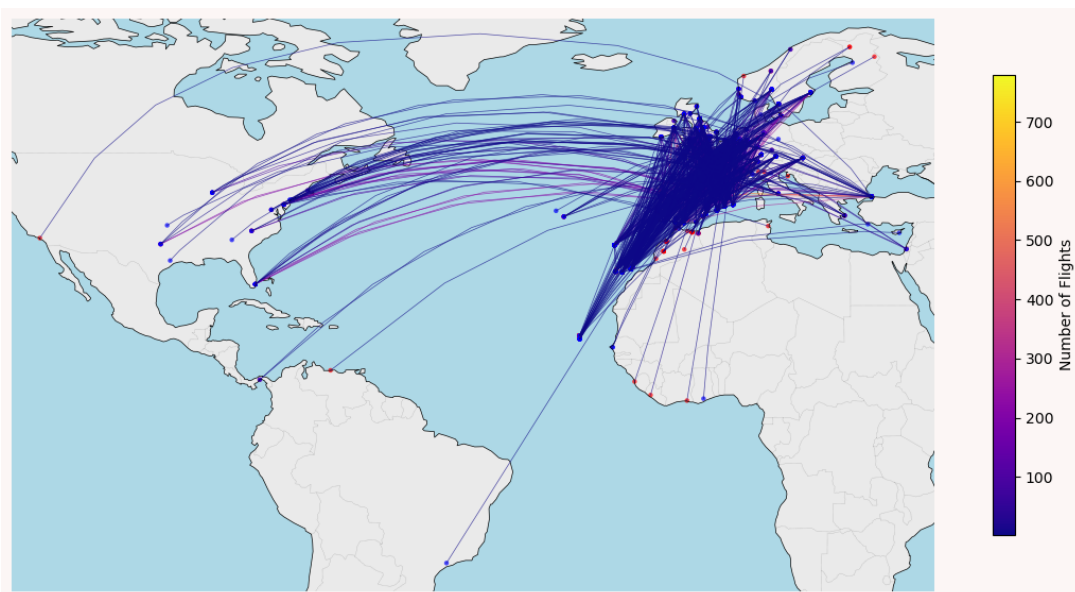


Figure 6.5: Origin-destination pairs and corresponding number of flight observations.

6.3. Exploratory Data Analysis

The dataset comprises flights operated by 17 airlines utilizing 25 different aircraft types, all of which fall under the medium or heavy wake turbulence categories. Figures 6.6 and 6.7 depict the distribution of flight operations by airline and aircraft type. Remarkably, nearly 80% of all flights were conducted by six primary airlines (TOM, SWR, BEL, SAS, JAF, and EIN), with TOM and SWR combining to represent 30% of the total operations. In terms of aircraft types, the distribution is even more pronounced, with over 50% of flights being operated by two specific aircraft types: A320 and B738.

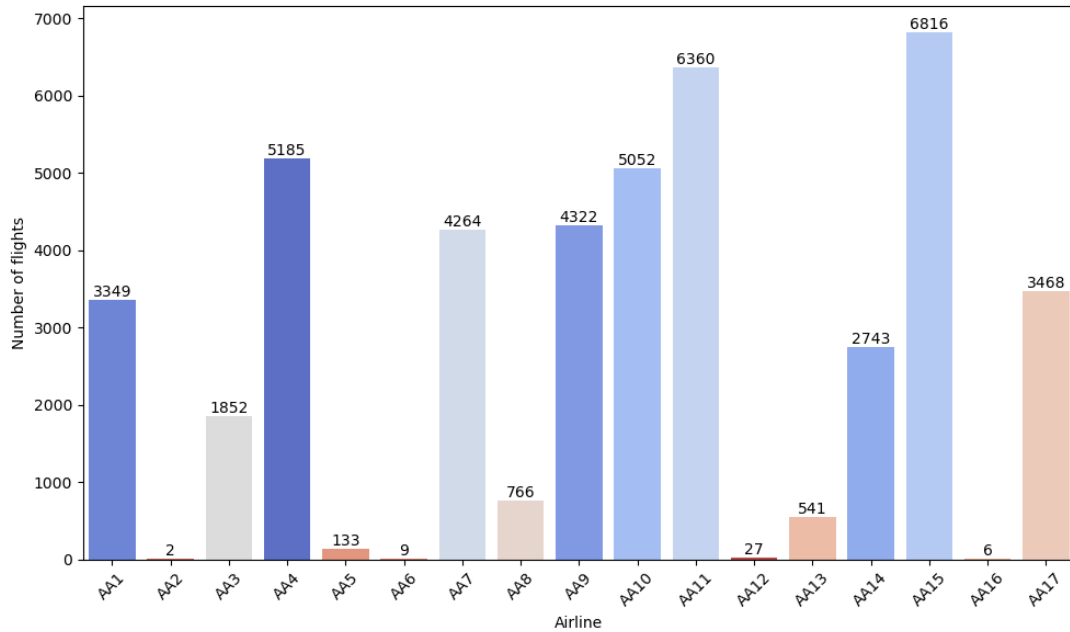


Figure 6.6: Distribution of flight traffic by airline.

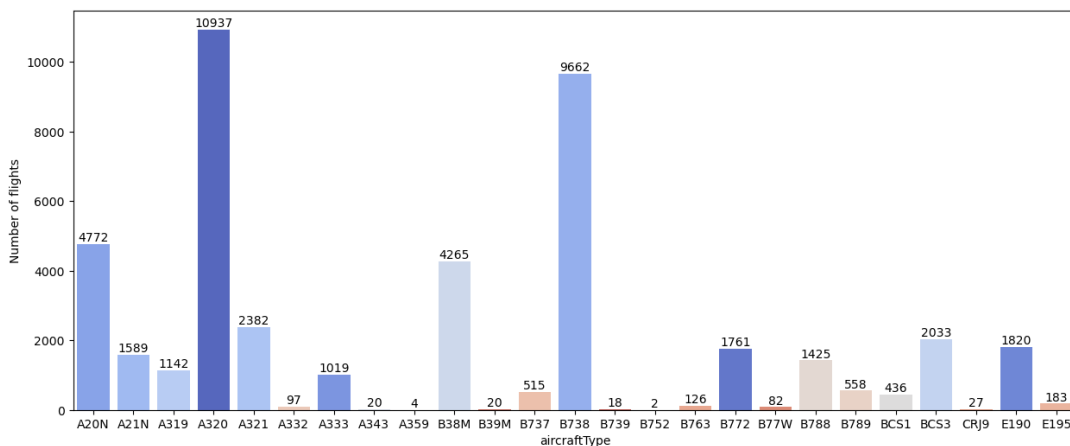


Figure 6.7: Distribution of flight traffic by aircraft type.

The distribution of the Wake Turbulence Category, represented by the `wake` feature in Figure 6.8, reveals a marked imbalance, characterized by a ratio of approximately 7.8 medium (M) flights for every heavy (H) flight. Given that the objective of this thesis is to predict TOW across the complete spectrum of possible aircraft weights or scenarios, a single model trained on this imbalanced dataset is likely to be skewed towards

the more prevalent TOW ranges associated with medium wake category flights. This issue will be explored in greater detail in Section 6.4.

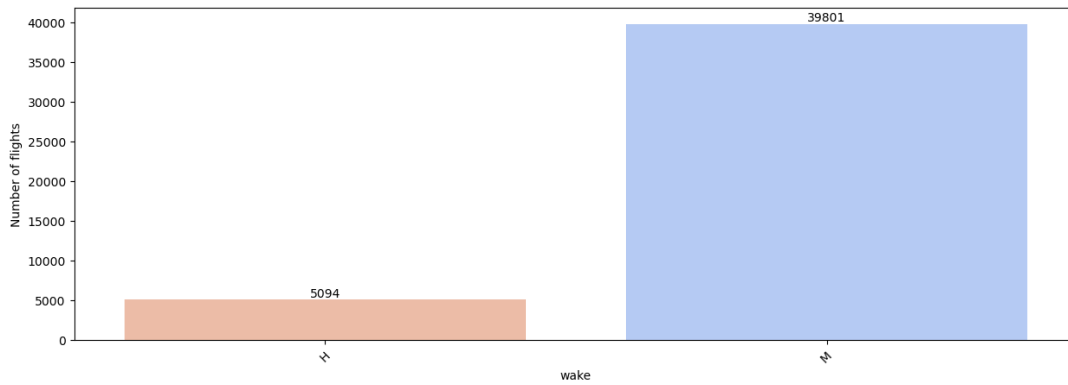


Figure 6.8: Distribution of Wake Turbulence Category.

The distributions shown in Figures 6.9, 6.10a, and 6.10 illustrate the distribution of TOW, overall flight duration, and total flight distance. The TOW distribution is notably skewed, with a pronounced peak observed in the range of 50,000 kg to 100,000 kg. This concentration of lower TOW values corresponds directly with the high prevalence of medium wake category aircraft, which typically reside within this weight interval. Although the distribution extends to higher TOWs (exceeding 300,000 kg), representing heavier aircraft, such occurrences are significantly less frequent, underscoring the scarcity of data for larger aircraft types.

The consistency of the dataset's skewness is further corroborated by the two additional distributions, which indicate that most flights are relatively short, peaking at durations of 2 to 3 hours, with a sharp decline in frequency for longer flights. Similarly, the majority of flights cover shorter distances, with a significant peak occurring between 500 and 1,000 nautical miles, followed by a rapid decrease for longer distances. Both the duration and distance distributions are heavily skewed towards lower values, reflecting the patterns observed in TOW and wake categories. This strong correlation among these variables support the conclusion that the dataset primarily comprises medium-sized aircraft engaged in short to medium operations.

Furthermore, the distribution of one the aircraft-related features, MTOW, as illustrated in Figure 6.11, reveals a multimodal and significantly imbalanced distribution. Distinct peaks are observable, particularly around 75,000 kg (10,935 flights) and 79,000 kg (9,662 flights). These peaks align closely with the MTOWs of the most common aircraft types represented in the dataset (e.g., A320, B738). Beyond these dominant categories, there is a notable drop-off in flight counts, with very few flights recorded for mid-range MTOWs (e.g., between 83,000 kg and 181,400 kg). While instances of higher MTOW categories do exist, their counts remain significantly lower (e.g., 1,426 flights at 217,000 kg and 82 flights at 351,120 kg), reaffirming the insufficient representation of heavy aircraft within the dataset. This intricate MTOW distribution highlights the challenge of generalizing predictive models across various aircraft sizes due to the uneven data representation.

The ratio distribution of TOW-to-MTOW, as shown in Figure 6.12, provides essential insights into the relationships between these two variables. Each vertical column of points represents the range of TOWs for flights sharing a specific MTOW. The plot

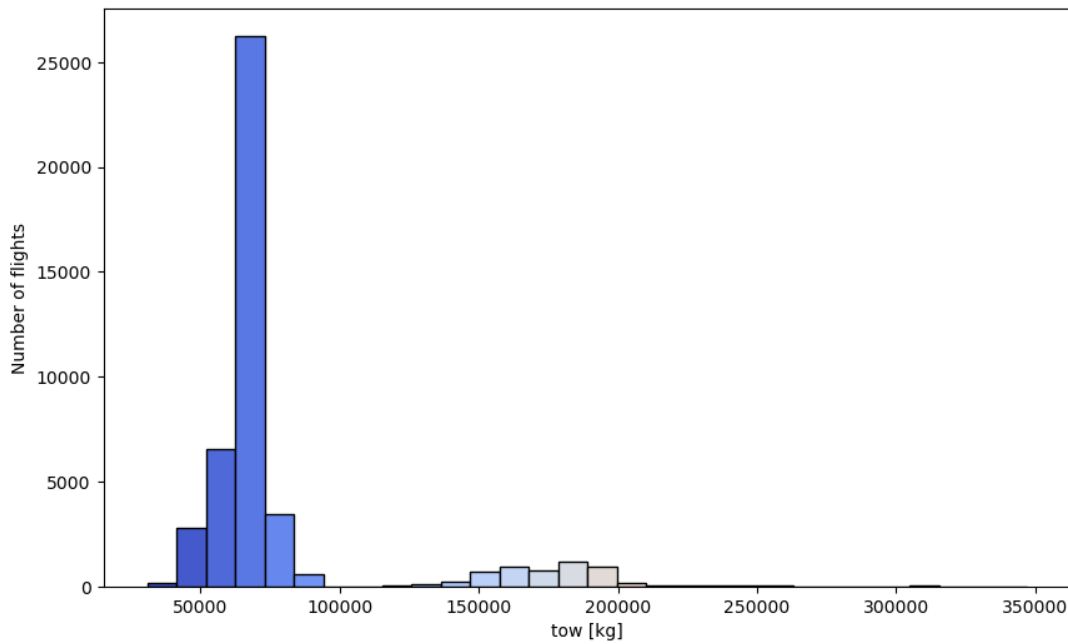
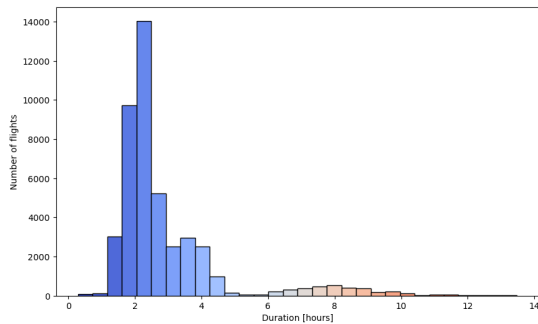
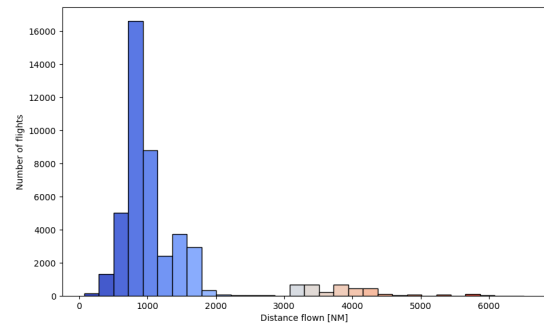


Figure 6.9: Distribution of TOW.



(a) Distribution of overall flight duration.



(b) Distribution of total covered distance.

Figure 6.10: Distribution of engineered features.

clearly shows that for any given MTOW, the actual TOW varies, indicating differing fuel and payload configurations. For aircraft with lower MTOWs, the actual TOWs often appear closer to their maximum capacity. As MTOW increases, particularly for larger aircraft, the absolute spread of actual TOWs tends to extend, meaning these aircraft operate with a greater range of payloads relative to their maximum capacity. The blue dashed lines indicate how visually the data is segmented and highlighting the notion to develop different models for medium and heavy wake category flights.

The distribution of TOW among the most prevalent aircraft types, as illustrated in Figure 6.13, facilitates a direct comparison of TOW characteristics across the following aircraft models: A320, B738, A20N, B38M, and A321. For each aircraft type, the analysis includes the median TOW, the interquartile range (IQR), and the dispersion of data points. Overlaying this information are the specific MTOW (represented by a red dot), MLW (represented by a blue dot), and OEW represented by a green dot) for

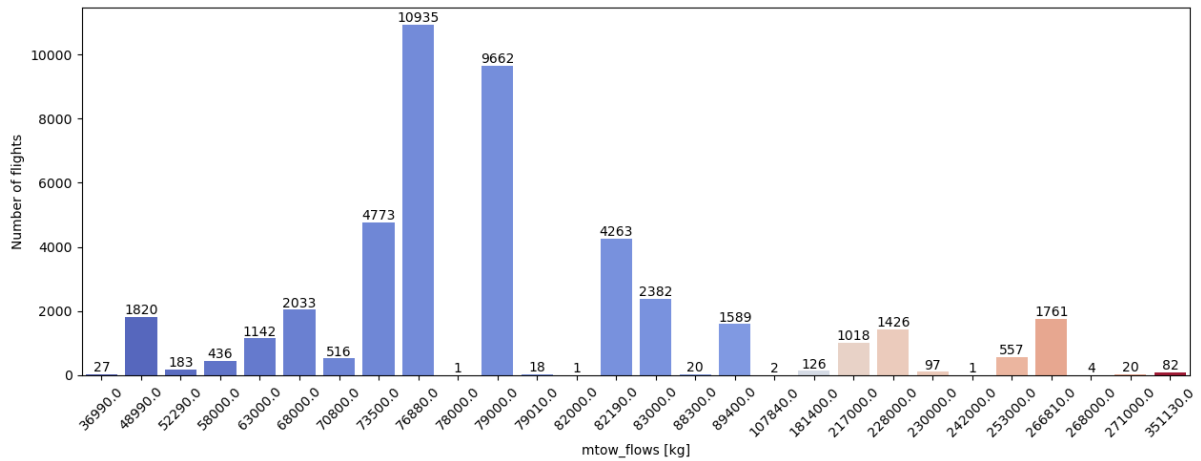


Figure 6.11: Distribution of MTOW.

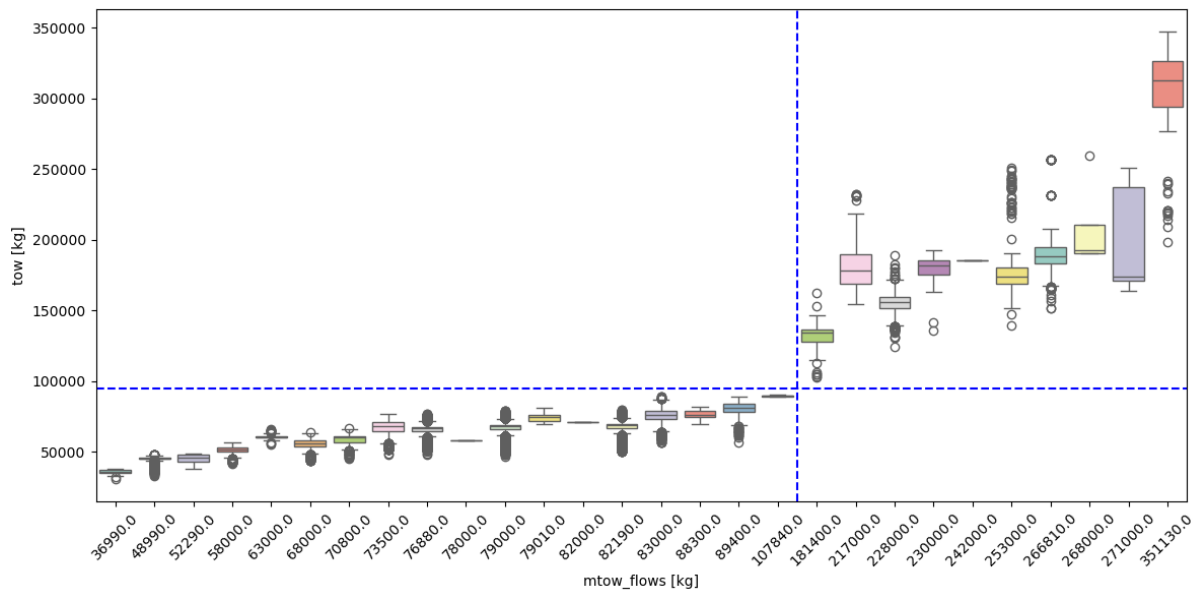


Figure 6.12: Ratio distribution of TOW-to-MTOW.

each model. It is important to highlight that, although all these aircraft are classified within the medium wake category and account for a significant portion of flight operations, their TOW distributions exhibit slight variations. The A321, identified as the largest variant among these models, demonstrates a marginally higher median TOW and a broader range in comparison to the others. Importantly, the observed TOW values for all types consistently lie between their respective OEW and MTOW, affirming the realism of the data. The identified outliers signify instances where aircraft were operated at the extreme limits of their operational weight envelope.

In summary, the exploratory data analysis provides valuable insights into the characteristics and patterns of the flight dataset. The monthly and hourly traffic distributions highlight significant trends in flight demand, influenced by seasonal and temporal factors. The analysis of spatial coverage confirms the dataset's robust representation of European air traffic. Furthermore, the identification of dominant air-

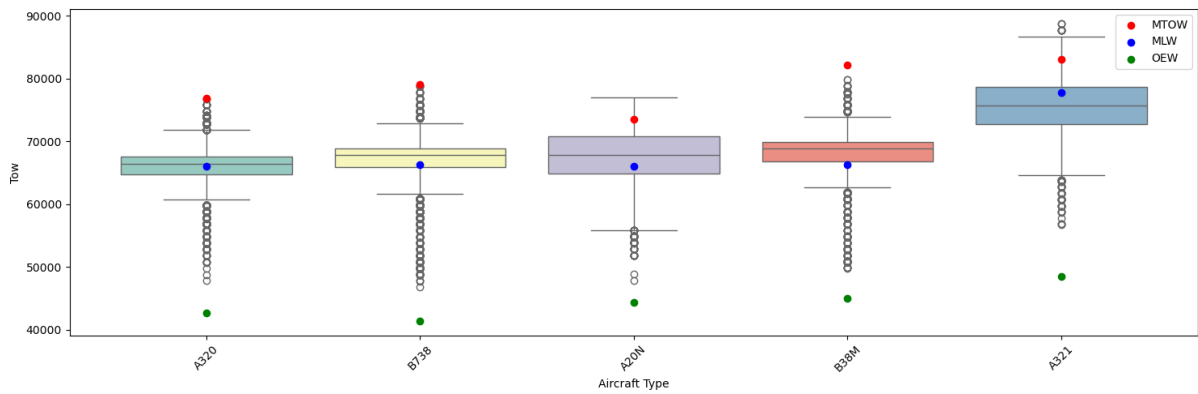


Figure 6.13: Distribution of TOW over the most common aircraft types.

lines and aircraft types underscores operational trends within the context of the industry. Notably, the observed imbalance in the wake category distribution presents challenges for predictive modelling, necessitating to address the skewness towards medium-weight aircraft. Overall, these findings cover the way for a more focused analysis of the factors affecting flight operations and performance.

6.4 Case Studies

To address the skewness observed in the wake turbulence category, three distinct case studies were undertaken. As previously demonstrated, the dataset is mainly composed of medium wake flights, which are more prevalent than heavy wake flights. Furthermore, there is a notable distinction in the variance of their respective MTOW values, overall flight durations, and total distances covered. For the purposes of enhancing the modelling development process, it was essential to separate the dataset by wake category. This separation will mitigate the high variability associated with heavy wake flights, as predictors for these flights do not significantly impact the prediction capabilities for medium wake flights, which are represented in a larger volume and exhibit normal variability in their predictors.

Additionally, it is important to consider that not all flights have associated climb-velocity features, depending on the availability of the Climb Phase Trajectory. As discussed in Section 6.2, imputing missing values for these variables can result in suboptimal data handling due to the high variability of speed features during the early climb phase. The flights lacking a Climb Phase Trajectory correspond to those landing or overflying the Spanish FIR. With these considerations in mind, the following case studies are presented:

- **Dataset with Climb Phase Trajectory and Medium (M) Wake Category:** This dataset comprises flights that take off from Spain and include aircraft categorized as medium wake.
- **Dataset without Climb Phase Trajectory and Medium (M) Wake Category:** This dataset serves as the complement to the previous one, containing flights that do not take off from Spain while still including aircraft within the medium wake category.
- **Dataset with Heavy (H) Wake Category:** This dataset encompasses all flights

associated with aircraft classified in the heavy wake category, regardless of their take-off locations. Notably, climb-velocity features are excluded from this dataset.

It is important to note that the three case studies effectively partition the dataset into scenarios where either data skewness or the unavailability of certain features is no longer an issue. The Dataset concerning the Heavy (H) Wake Category was not separated into datasets based on the presence of the Climb Phase Trajectory, due to the limited number of flights (5094) associated with the heavy wake category. Reducing the dataset size further could lead to an increased risk of overfitting. Consequently, the three case studies outlined will serve as the foundation for the development of the ML model pipeline to predicting TOW.

6.5 Predictive Models

The selected GBDT algorithms for this project include **XGBoost**, **CatBoost**, **LightGBM**, and the **H2O AutoML** module. These cutting-edge algorithms will be integrated into a comprehensive pipeline designed to facilitate the automated training and evaluation of models for TOW prediction. The pipeline consists of several key components, each playing a crucial role in streamlining the entire process:

1. **Data Management:** This initial phase is responsible for loading the case study datasets, which are categorized based on the availability of climb trajectory data. Specifically, it delineates flights with documented climb phase trajectories from those lacking this information, as well as distinguishing between wake categories, medium and heavy.
2. **Hyperparameter Optimization:** This component leverages Optuna to conduct Bayesian optimization aimed at enhancing model performance. With a structured approach, it sets a 6-hour timeout limit and permits up to 20 trials. This methodical exploration of hyperparameter space allows the pipeline to identify the most effective model configurations, driving improved accuracy and efficiency in TOW predictions.
3. **Model Training:** The training phase incorporates several best practices to optimize model performance and ensure robust evaluation. The dataset is partitioned using an 80-20 train-test split, providing sufficient training data while maintaining an adequate test set for unbiased performance assessment. To prevent overfitting and enhance generalization capability, early stopping mechanisms are implemented during training, monitoring validation loss to halt training when performance plateaus. Feature selection techniques are systematically applied to identify the most informative variables, reducing computational complexity while maintaining predictive accuracy. Model performance is evaluated using a comprehensive suite of metrics: R^2 , MAE, RMSE, and MAPE, that will be detailed in Section 6.4. This multi-metric evaluation framework ensures thorough assessment of model accuracy, reliability, and practical applicability across different operational scenarios.
4. **Experiment Tracking:** To facilitate effective management of models and their corresponding experiments, the pipeline integrates MLflow. This powerful tool offers comprehensive logging capabilities, enabling the team to meticulously

track changes, compare model performance across different experiments, and maintain a clear record of all relevant parameters and outcomes.

5. **Visualization:** An integral aspect of the pipeline is its ability to generate automated visualizations. This includes detailed plots for residuals analysis, which helps diagnose model performance issues, as well as representations of feature importance, providing insights into the factors most influential in the predictions. Performance evaluation plots further enhance understanding of model behavior and effectiveness.
6. **Model Persistence:** Finally, the trained models are stored efficiently using binary serialization methods along with H2O's native format. This ensures that models can be easily retrieved and deployed for future use, facilitating the implementation of developed models in operational scenarios.

6.5.1 XGBoost

XGBoost has been implemented with hyperparameter tuning facilitated through Optuna's Tree-structured Parzen Estimator (TPE) algorithm. The model incorporates early stopping with a patience of 50 rounds and is evaluated using the RMSE metric, which is particularly effective in penalizing larger errors more heavily compared to other testing metrics. Table 6.1 presents all the hyperparameters that were tuned along with their corresponding ranges.

For the **Learning Rate** hyperparameter, a logarithmic scale is selected due to the multiplicative effects of the learning rate on the model's performance. This range achieves a balance between training stability, where the lower bound prevents overfitting, and convergence speed, the upper bound ensuring a reasonable training duration. Values below 1e-3 may result in prolonged training times, while those above 0.1 often lead to instability.

The selected range for **Max Depth** hyperparameter is designed to mitigate both underfitting (depth < 3) and overfitting (depth > 10). Estimating aircraft weight typically involves moderately complex feature interactions that do not necessitate excessively deep trees. A linear scale is appropriate here, as the effects of depth are also linear. Concerning the **Min Child Weight** hyperparameter, this regularization parameter governs the minimum sum of instance weights in child nodes. The chosen range support adequate regularization (minimum of 1) while avoiding excessive regularization (maximum of 10), which could hinder the model from learning significant patterns in flight data.

For the **Gamma** hyperparameter, the minimum threshold for loss reduction is specified on a logarithmic scale since its effect is multiplicative concerning the loss function. Smaller values (1e-3) allow for the majority of splits, whereas larger values (1.0) provide substantial regularization. The range for **Subsample** ensures that a sufficient amount of data is utilized for learning (minimum at 50%) while permitting full dataset usage (100%).

In regard to **Colsample_bytree** hyperparameter, the established range maintains adequate feature representation (40%) while also allowing for complete utilization of features, thus helping to prevent overfitting to specific characteristics in the flight data. The parameters **Reg_alpha** and **Reg_lambda** are employed for L1 and L2 regular-

Methodology

ization respectively, using a logarithmic scale due to their multiplicative influences on the objective function. This range accommodates minimal regularization (1e-3) to strong regularization (10.0), addressing various complexities inherent within the dataset.

Table 6.1: Hyperparameters of the XGBoost model and the corresponding ranges utilized in Optuna.

Hyperparameter	Range	Type	Log Scale	Description
learning_rate	[1e-3, 0.1]	float	Yes	Step size shrinkage aimed at preventing overfitting.
max_depth	[3, 10]	int	No	Maximum depth of individual trees.
min_child_weight	[1, 10]	int	No	Minimum sum of instance weights in child nodes.
gamma	[1e-3, 1.0]	float	Yes	Minimum loss reduction required for a split.
subsample	[0.5, 1.0]	float	No	Fraction of samples utilized per tree.
colsample_bytree	[0.4, 1.0]	float	No	Fraction of features employed per tree.
reg_alpha	[1e-3, 10.0]	float	Yes	L1 regularization term applied to weights.
reg_lambda	[1e-3, 10.0]	float	Yes	L2 regularization term applied to weights.

6.5.2 LightGBM

LightGBM operates utilizing a leaf-wise tree growth strategy, which significantly enhances efficiency for large datasets. The implementation includes features for handling categorical data as well as employing gradient-based one-side sampling (GOSS). Table 6.2 illustrates all the hyperparameters that were tuned alongside their corresponding ranges.

For the **Learning Rate** hyperparameter, similar to XGBoost, a logarithmic scale effectively reflects the multiplicative nature of its effects. This range seeks to achieve a balance between convergence speed and training stability. The **Num Leaves** hyperparameter is bounded by a minimum of 20, ensuring sufficient model complexity, while the maximum of 150 prevents overfitting. A linear scale is suitable in this context, as the effects of leaves are additive.

The range for **Min Child Samples** hyperparameter ensures that sample size is statistically significant (minimum of 5) while also preventing over-regularization (maximum of 50), which could adversely affect model generalization. Although the range for **Min Child Weight** hyperparameter is quite extensive, its diminishing returns justify a linear scale. The minimum is kept small to provide flexibility, while the upper bound offers appropriate regularization. In terms of **Subsample** hyperparameter, as with XGBoost, this parameter guarantees adequate data representation while also permitting full utilization of the dataset. Regarding **Colsample_bytree** hyperparameter, the

established range promotes feature diversity, thereby preventing overfitting to specific flight parameters. The hyperparameters **Reg_alpha** and **Reg_lambda** serve as regularization terms, both employing logarithmic scaling due to their multiplicative impact on the loss function.

Table 6.2: Hyperparameters of the LightGBM model and the corresponding ranges utilized in Optuna.

Hyperparameter	Range	Type	Log Scale	Description
learning_rate	[1e-3, 0.1]	float	Yes	Step size shrinkage to mitigate overfitting.
num_leaves	[20, 150]	int	No	Number of leaves in the tree.
min_child_samples	[5, 50]	int	No	Minimum number of data points required in a child node.
min_child_weight	[1e-3, 10]	float	No	Minimum sum of instance weights in child nodes.
subsample	[0.5, 1.0]	float	No	Fraction of data utilized for each tree.
colsample_bytree	[0.4, 1.0]	float	No	Fraction of features utilized for each tree.
reg_alpha	[1e-3, 10.0]	float	Yes	L1 regularization term on weights.
reg_lambda	[1e-3, 10.0]	float	Yes	L2 regularization term on weights.

6.5.3 CatBoost

CatBoost is a highly effective GBDT library specifically designed to handle categorical features directly, eliminating the need for extensive preprocessing. Its innovative use of ordered boosting techniques mitigates the risk of overfitting, enhancing model robustness and reliability. The implementation includes automatic detection of categorical features and provides built-in metrics for feature importance, allowing users to gain insights into the contribution of each feature to the model's predictions.

Table 6.3 summarizes the hyperparameters that were tuned, along with their corresponding ranges. The selection of a logarithmic scale for the **Learning Rate** hyperparameter acknowledges the multiplicative impact of learning rates on the convergence behaviour of the model. For the **Depth** hyperparameter, the specified range balances the complexity of the model against the potential risks of overfitting, ensuring that the model is neither too simplistic nor overly complex. The **Random Strength** parameter, also represented on a logarithmic scale, indicates the degree of randomness incorporated during the scoring of splits. Increased values typically enhance model robustness but may inadvertently reduce the precision of predictions.

Regarding **Colsample_bylevel**, the wide-ranging interval (5% to 100%) facilitates aggressive feature selection while still conserving model capability. The **Min Data in Leaf** hyperparameter, similar to the `min_child_samples` in LightGBM, ensures that leaves within the trees maintain statistical significance, thereby preventing excessive

Methodology

regularization that could hinder model performance. Lastly, the **Leaf Estimation Iterations** hyperparameter set a balance between computational efficiency and accuracy; while higher values enhance the accuracy of leaf value estimates, they understandably require additional training time.

Table 6.3: CatBoost hyperparameters and corresponding ranges used in Optuna.

Hyperparameter	Range	Type	Log Scale	Description
learning_rate	[1e-3, 0.1]	float	True	Step size shrinkage to prevent overfitting.
depth	[3, 10]	int	No	Maximum depth of individual trees.
random_strength	[10.0, 50.0]	float	Yes	Randomness for scoring splits.
colsample_bylevel	[0.05, 1.0]	float	No	Fraction of features used per tree level.
min_data_in_leaf	[5, 50]	int	No	Minimum samples required in a leaf.
leaf_estimation_iterations	[1, 15]	int	No	Number of iterations for leaf value estimation.

The selected hyperparameter ranges are intentionally broad, empowering the Optuna optimization framework to effectively explore the hyperparameter landscape. This exploration is critical for achieving an appropriate balance between underfitting and overfitting, given the unique characteristics of each dataset encountered in case studies. Utilizing logarithmic scales for certain parameters (specifically, `learning_rate`, `gamma`, `reg_alpha`, `reg_lambda`, and `random_strength`) follows best practices, facilitating a more efficient search when parameter effects span multiple orders of magnitude.

6.5.4 Evaluation Metrics

The performance of the developed prediction models is assessed using a comprehensive suite of evaluation metrics, each providing unique insights into different aspects of model accuracy and reliability. This multi-metric approach ensures thorough evaluation of model performance across various dimensions, enabling robust comparison between different algorithms and configurations.

6.5.4.1 Coefficient of Determination

The coefficient of determination, denoted as R^2 , quantifies the proportion of variance in the dependent variable that is predictable from the independent variables. Mathematically expressed as:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \quad (6.5)$$

where SS_{res} represents the sum of squares of residuals and SS_{tot} represents the total sum of squares. R^2 values range from 0 to 1, with values closer to 1 indicating better model fit. In the context of TOW prediction, R^2 provides insight into how well the

model captures the underlying relationships between flight parameters and aircraft weight. A high R^2 value indicates that the model successfully explains the majority of variance in TOW, suggesting strong predictive capability and good feature selection.

6.5.4.2 Mean Absolute Error (MAE)

Mean Absolute Error represents the average absolute difference between predicted and actual values, calculated as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (6.6)$$

where n is the number of observations, y_i represents actual values, and $\hat{y}_i$ represents predicted values. MAE provides an intuitive measure of prediction accuracy in the same units as the target variable (kg). This metric is particularly valuable for operational applications as it directly indicates the expected magnitude of prediction errors. MAE is robust to outliers compared to squared error metrics, making it suitable for datasets with occasional extreme values. In aeronautics contexts, MAE allows stakeholders to understand the typical prediction error magnitude for operational planning and safety margin calculations.

6.5.4.3 Root Mean Squared Error (RMSE)

Root Mean Squared Error is calculated as the square root of the mean of squared differences between predicted and actual values:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (6.7)$$

RMSE penalizes larger errors more heavily than MAE due to the squaring operation, making it more sensitive to outliers and extreme prediction errors. This characteristic is particularly important for safety-critical applications where large prediction errors could have significant operational consequences. RMSE is expressed in the same units as the target variable, facilitating direct interpretation. In TOW prediction, RMSE provides insight into the model's ability to avoid significant prediction errors, which is crucial for maintaining operational safety margins and regulatory compliance.

6.5.4.4 Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error expresses prediction accuracy as a percentage, calculated as:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (6.8)$$

MAPE provides a scale-independent measure of prediction accuracy, making it particularly useful for comparing model performance across different aircraft types or

Methodology

weight ranges. This metric is especially valuable when dealing with TOWs that vary significantly across different aircraft categories, as it normalizes errors relative to the actual values. MAPE facilitates intuitive interpretation of model performance in percentage terms, allowing stakeholders to quickly assess whether prediction errors are within acceptable operational tolerances.

6.5.5 MLFlow Tracking

To ensure reproducibility, transparency, and systematic evaluation of the developed models, MLFlow has been employed to track hyperparameters, evaluation metrics, and artifacts associated with each model. The MLFlow dashboard, illustrated in Figure 6.14, offers a comprehensive interface for experiment tracking across all model runs within the project. This interface is presented in a tabular format with robust filtering capabilities, enabling users to search based on specific criteria, sort results by creation time, state, or datasets (case studies), and group runs according to datasets. The naming convention for the runs consists of a unique identifier that combines the model type name and a timestamp. Furthermore, the interface includes controls for time filtering, active state management, and dataset selection, showcasing MLFlow's versatile experiment management functionalities.

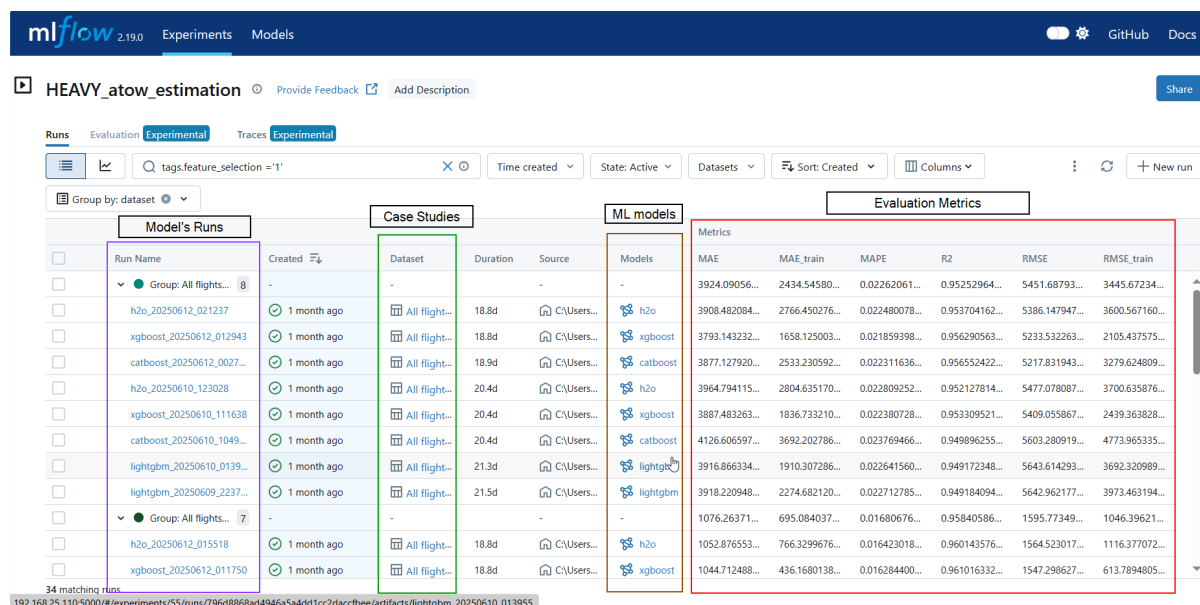


Figure 6.14: MLFlow Dashboard elements.

With regard to parameter and metrics logging, Figure 6.15 provides an example of a run designated as `h2o_20250610_123028`, which is associated with the **Dataset with Heavy (H) Wake Category**. This particular run tracked 65 parameters relevant to the H2O model. Additionally, it includes four key performance metrics for a thorough evaluation of the model: MAE, MAPE, R^2 , and RMSE. Moreover, two training validation metrics: MAE_train and RMSE_train are critical for diagnosing common model issues such as overfitting, a topic which will be explored in detail in Chapter 7.

Furthermore, the capabilities for artifact logging are depicted in Figure 6.16, which illustrates the file structure maintained by MLFlow. The system archives a variety of

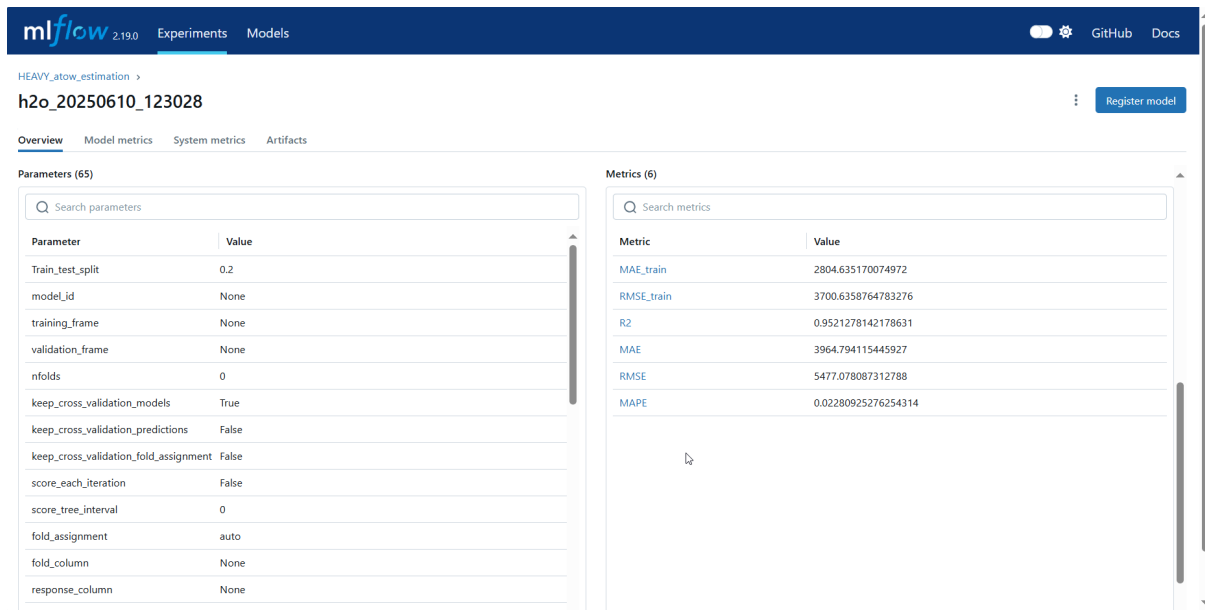


Figure 6.15: MLflow model's parameters and metrics.

visualization artifacts, including residuals distribution plots, scatter plots, and predictions versus residuals plots. Specifically for H2O, the comprehensive leaderboard featuring all models tested, ranked by the RMSE metric, is stored as an artifact in CSV format. The artifacts section encompasses model-specific files such as the model binary, along with essential configuration files including `h2o.yaml`, `conda.yaml`, and `python_env.yaml` for environment specifications, as well as `requirements.txt` for dependency management.

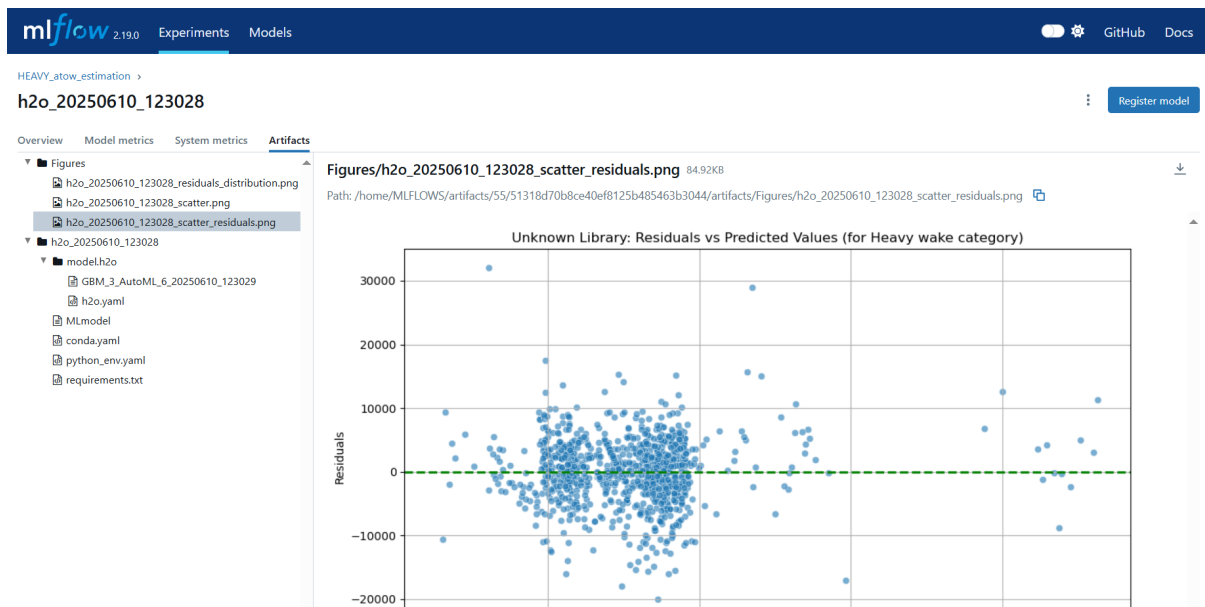


Figure 6.16: MLflow Artifacts.

Chapter 7

Results

This section presents the results of the development of the prediction models, starting with the analysis and selection of the most optimal model for each of the case studies from Section 7.1 to Section 7.3. Section 7.4 discusses the applicability of the best-performing models, as described in Chapter 6.

7.1 Flights with Climb Phase Trajectory and Medium (M) Wake Category Case

Of the two approaches proposed in Chapter 6, about training fine-tuned GBDT algorithms and to leverage the H2O framework to automatically train and tune a variety of models. H2O AutoML outperformed the approach of the fine-tuned GBDT algorithms. For this case, the best H2O model got a RMSE of 1297.839, a MAE of 884.389, a MAE_train of 597.737, a MAPE of 0.013, and a R^2 of 0.968. This model not only got the lowest errors metrics and high variance explainability, but also the gap between MAE and MAE_train is the smallest (286.65) for all the trained models, maintaining the lowest potential overfitting.

Figures 7.1, 7.2, and 7.3, shows a scatter plot between the predicted values and the actual values, a distribution of the residuals, and a scatter plot between residuals and predicted values. The spread of the points around the diagonal line in 7.1 represents the model's errors. The red points, suggest that the model might be struggling to predict some atypical values as accurately as the rest. And the horizontal patterns around the diagonal line may indicate that actual tow values are likely to be estimated (discretized) for the samples in these patterns.

Figure 7.2 illustrates the distribution of the residuals which appears to be approximately normal and centred around zero. The few atypical values from before, have associated extreme residuals values. Although the distribution seems somewhat narrow and peaked, this suggests the model residuals are generally well-behaved but may have some minor deviations from perfect normality.

Figure 7.3 shows residuals against predicted values, suggesting a well-performing regression model with residuals that are approximately normally distributed and randomly scattered, aside of from the linear patterns that are cause of how actual tow was estimated at the beginning.

7.1. Flights with Climb Phase Trajectory and Medium (M) Wake Category Case

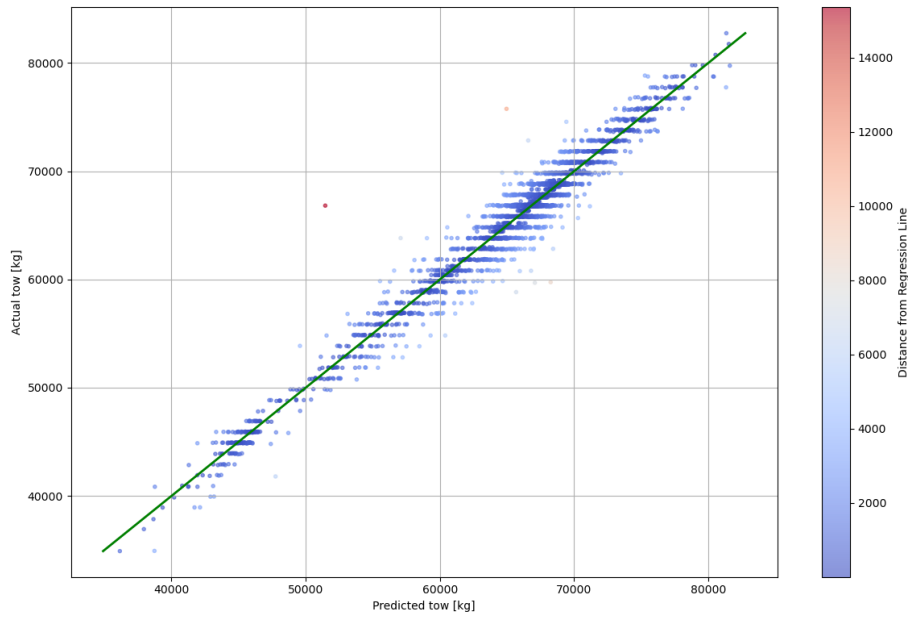


Figure 7.1: Regression Overview of Case Study: Flights with Climb Phase Trajectory and Medium (M) Wake Category.

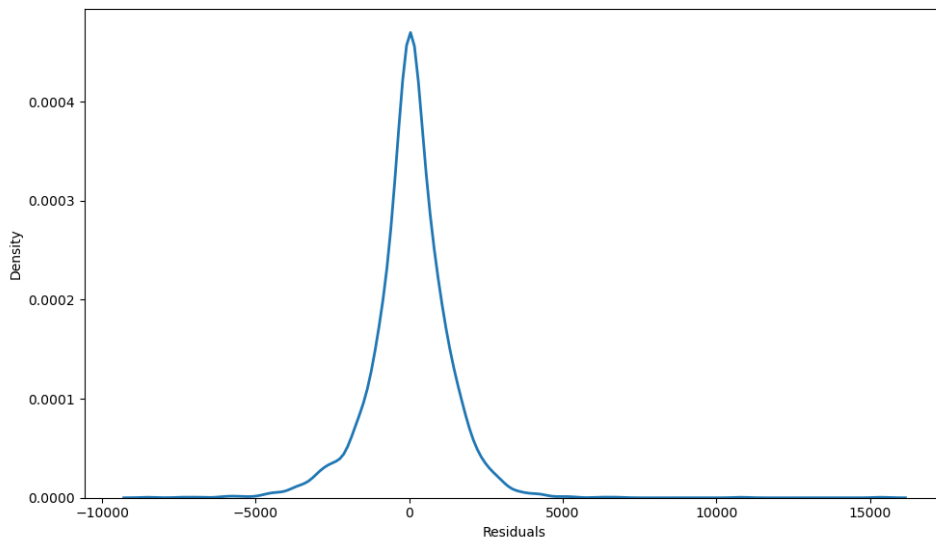


Figure 7.2: Residuals Distribution of Case Study: Flights with Climb Phase Trajectory and Medium (M) Wake Category.

7.1.1 Results Analysis

The results of Table 7.1 showed a good predictive performance across all models, particularly due to the inclusion of features derived from the climb phase. XGBoost stood out slightly with an RMSE of 1333.135, an MAE of 914.43, and an R^2 of 0.967, closely followed by CatBoost and LightGBM, both with the similar MAE and MAPE values (0.014).

Notably the gap between MAE and MAE_train is the smallest for CatBoost (315.14), indicating only a bit generalization gap and a very low sign of overfitting. The other

Results

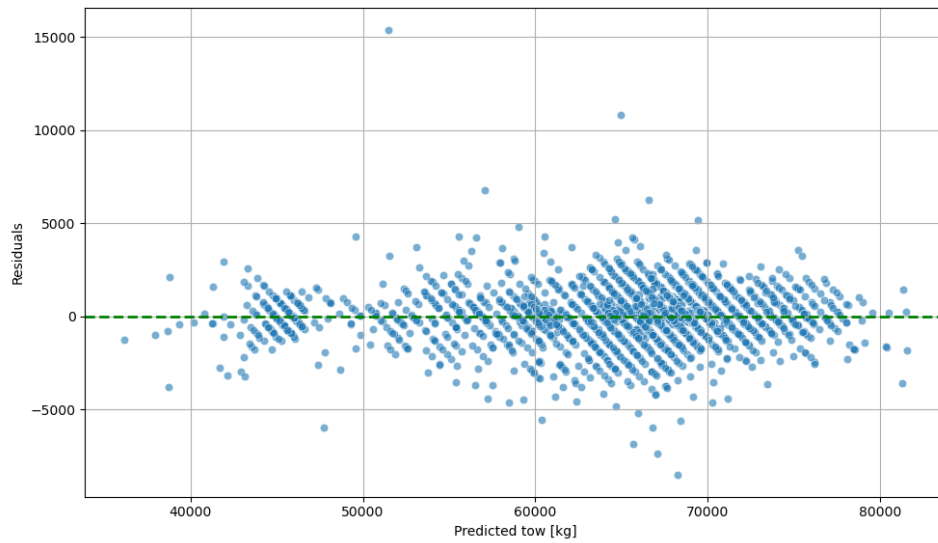


Figure 7.3: Residuals against Predictions Scatter plot of Case Study: Flights with Climb Phase Trajectory and Medium (M) Wake Category.

two models are more likely to present overfitting, because they have a gap of 513.40 and 600.39 for LightGBM and CatBoost respectively.

This positions CatBoost as the more suitable model for this case, for its low error values and generalization capability. The inclusion of climb phase trajectory data, capturing early vertical motion, rate of climb, and early flight dynamics, proved an enhance in TOW predictions.

Figure 7.4 presents the top 20 feature importances obtained from the trained CatBoost model. These importances were computed using the `PredictionValuesChange` method [57], indicating the average change in prediction if a feature's value is altered. As expected, aircraft features are among the most critical predictors overall. It is also significant the contribution of the physics-based features `CLIMB_vel_z_median` and `CRUISE_modo_c_median` describing the rate of climb and altitude during cruise phase.

7.2 Flights without Climb Phase Trajectory and Medium (M) Wake Category Case

For this case, the best H2O model got a RMSE of 1748.455 , a MAE of 1140.858 , a MAE_train of 780.339 , a MAPE of 0.017 , and a R2 of 0.954 . This model not only got the lowest errors metrics and high variance explainability, but also the gap between MAE and MAE_train is the smallest (360.52) for all the trained models, maintaining the lowest potential overfitting.

Figures 7.5, 7.6, and 7.7, shows a scatter plot between the predicted values and the actual values, a distribution of the residuals, and a scatter plot between residuals and predicted values. Now the presence of orange and red points is greater than before presenting a greater sparsity around the diagonal line in 7.5 indicating that there are more predictions with higher errors. The same horizontal patterns than

7.2. Flights without Climb Phase Trajectory and Medium (M) Wake Category Case

Table 7.1: Comparison of model performance and hyperparameters for Flights with Climb Phase Trajectory and Medium (M) Wake Category.

Algorithm	Best Hyperparameters	RMSE	MAE	MAE_train	MAPE	R ²
XGBoost	learning_rate=0.022, max_depth=6, min_child_weight=9, gamma=0.168, subsample=0.650, colsample_bytree=0.909, reg_alpha=0.017, reg_lambda=0.001	1333.135	914.428	314.029	0.014	0.967
LightGBM	learning_rate=0.070, num_leaves=54, min_child_samples=38, min_child_weight=10, subsample=0.830, colsample_bytree=0.405, reg_alpha=6.067, reg_lambda=0.051	1377.532	930.599	417.202	0.014	0.964
CatBoost	learning_rate=0.031, depth=5, random_strength=47.36, colsample_bylevel=0.679, min_data_in_leaf=6, leaf_estimation_iterations=11	1337.789	929.114	613.977	0.014	0.966

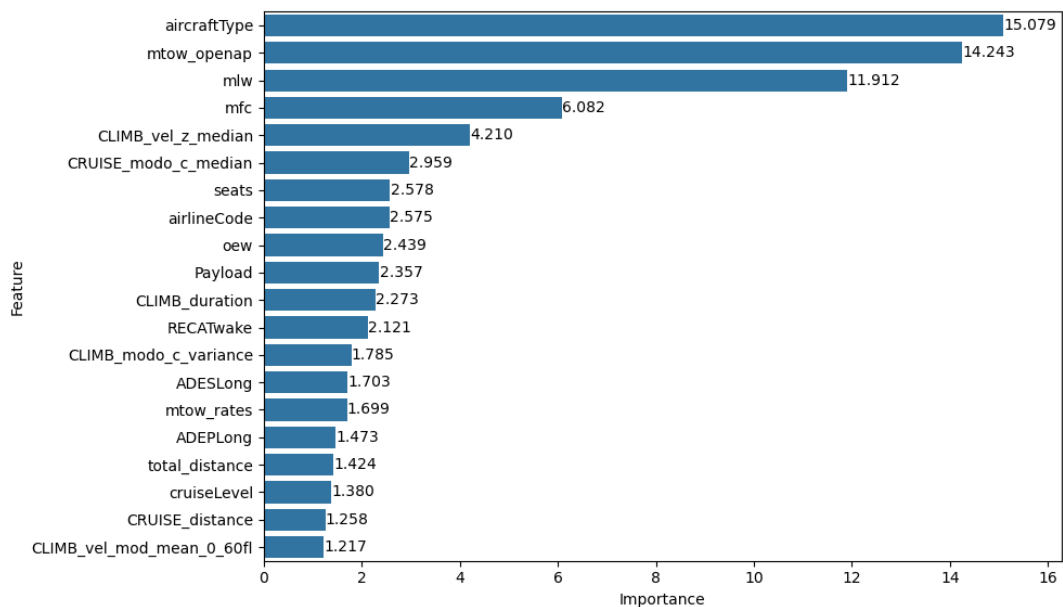


Figure 7.4: Top 20 feature importances for CatBoost of Case Study: Flights with Climb Phase Trajectory and Medium (M) Wake Category.

before present in a similar way an reinforce the idea that actual tow values for some samples were estimated.

Results

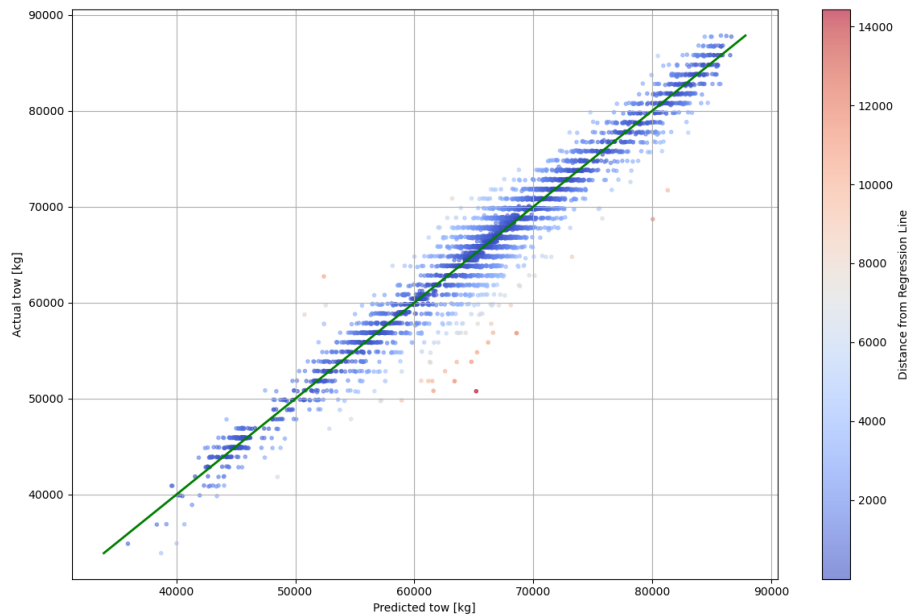


Figure 7.5: Regression Overview of Case Study: Flights without Climb Phase Trajectory and Medium (M) Wake Category.

Figure 7.6 illustrates the distribution of the residuals which appears to be approximately normal and centred around zero. The few atypical values have associated extreme residuals values. Although the distribution seems somewhat narrow and peaked, this suggests the model residuals are also generally well-behaved but may have some minor deviations from perfect normality.

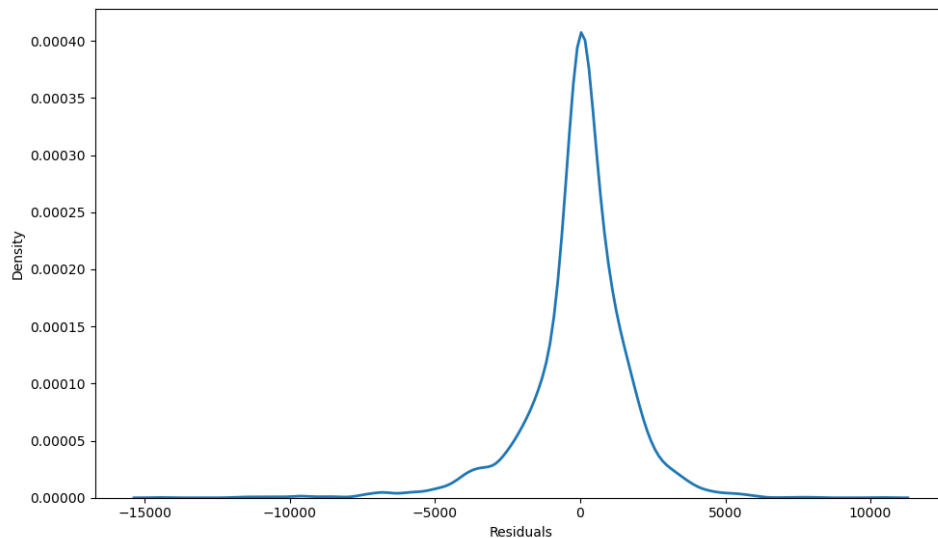


Figure 7.6: Residuals Distribution of Case Study: Flights without Climb Phase Trajectory and Medium (M) Wake Category.

Figure 7.7 shows residuals against predicted values, suggesting a well-performing regression model with residuals that are approximately normally distributed and randomly scattered, aside of from the linear patterns that are cause of how actual tow was estimated at the beginning.

7.3. Flights with Heavy (H) Wake Category Case

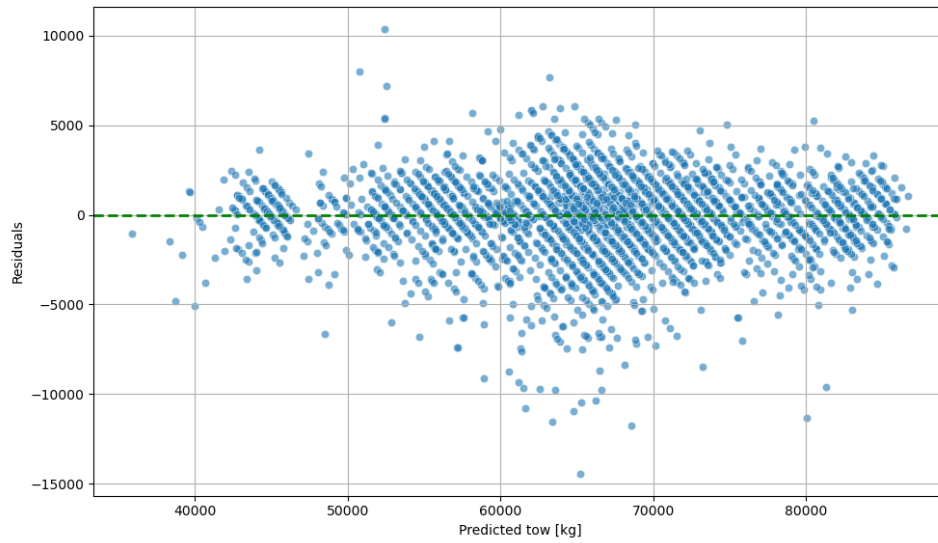


Figure 7.7: Residuals against Predictions Scatter plot of Case Study: Flights without Climb Phase Trajectory and Medium (M) Wake Category.

7.2.1 Results Analysis

The results of table 7.2 showed a modest drop in predictive performance by the lack of Climb Phase Trajectory. The RMSEs increased to around 1700 - 1800 kg, with CatBoost achieving the best RMSE of 1729.56, followed by XGBoost at 1755.71, and LightGBM at 1815.03. XGBoost retained the highest R^2 score at 0.953, indicating it still captured more variance despite the slightly higher error. All models still maintained reasonable accuracy, with MAPE values around (0.017-0.018).

Notably the gap between MAE and MAE_train is the smallest for XGBoost (404.08), indicating only a bit generalization gap and a very low sign of overfitting. LightGBM follows with a gap of 473.73, indicating moderate, and then CatBoost with a gap of 552.62, both indicating moderate overfitting.

This positions XGBoost as the more suitable model for this case, with an overall performance lower than in Section 7.1, due to the absence of the climb phase trajectory features, which reduced the model's ability to capture dynamic relationships between aircraft motion and TOW. Despite this limitation, the models maintained robust generalization capacity.

Figure 7.8 presents the top 20 feature importances obtained from the trained CatBoost model. Aircraft features again are among the most critical predictors overall. Highlighting the contribution of the feature `Payload` which in the absence of physics-based features, emphasizing the impact of operational loading without account with fuel load.

7.3 Flights with Heavy (H) Wake Category Case

For this case, the best H2O model got a RMSE of 5386.147, a MAE of 3908.482, a MAE_train of 3600.567, a MAPE of 0.022, and a R^2 of 0.953. This model not only got the lowest errors metrics and high variance explainability, but also the gap between

Results

Table 7.2: Comparison of model performance and hyperparameters for Flights without Climb Phase Trajectory and Medium (M) Wake Category.

Algorithm	Best Hyperparameters	RMSE	MAE	MAE_train	MAPE	R ²
XGBoost	learning_rate=0.002, max_depth=7, min_child_weight=3, gamma=0.628, subsample=0.760, colsample_bytree=0.550, reg_alpha=0.057, reg_lambda=9.257	1755.712	1161.933	757.848	0.018	0.953
LightGBM	learning_rate=0.005, num_leaves=122, min_child_samples=28, min_child_weight=8, subsample=0.718, colsample_bytree=0.541, reg_alpha=0.022, reg_lambda=0.033	1815.032	1115.732	642.000	0.017	0.950
CatBoost	learning_rate=0.032, depth=8, random_strength=13.109, colsample_bylevel=0.227, min_data_in_leaf=25, leaf_estimation_iterations=15	1729.555	1149.752	597.133	0.017	0.955

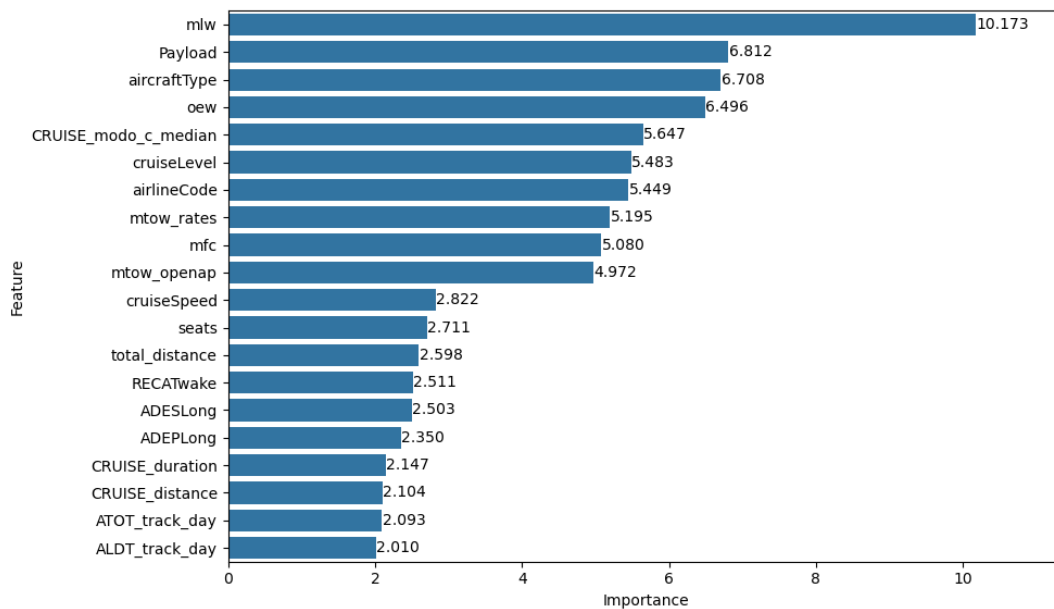


Figure 7.8: Top 20 feature importances for CatBoost of Case Study: Flights without Climb Phase Trajectory and Medium (M) Wake Category.

MAE and MAE_train is the smallest (1142.03) for all the trained models, maintaining a moderate potential overfitting.

7.3. Flights with Heavy (H) Wake Category Case

Figures 7.9, 7.10, and 7.11, shows a scatter plot between the predicted values and the actual values, a distribution of the residuals, and a scatter plot between residuals and predicted values. In Figure 7.9, the model seems to perform quite well in the middle range of t_{ow} between 150,000 - 200,000 where the density of the points is highest and has associated a low error.

For other side, there appears to be a more spread and a higher proportion of red and orange points at the higher end of the t_{ow} values (above 200,000) and to some extent at the lower end (below 150,000). This suggests that the model might be less accurate when predicting very high or very low t_{ow} values. In this case, this is due to fewer training samples, that prevent the model to learn the underlying patterns in data.

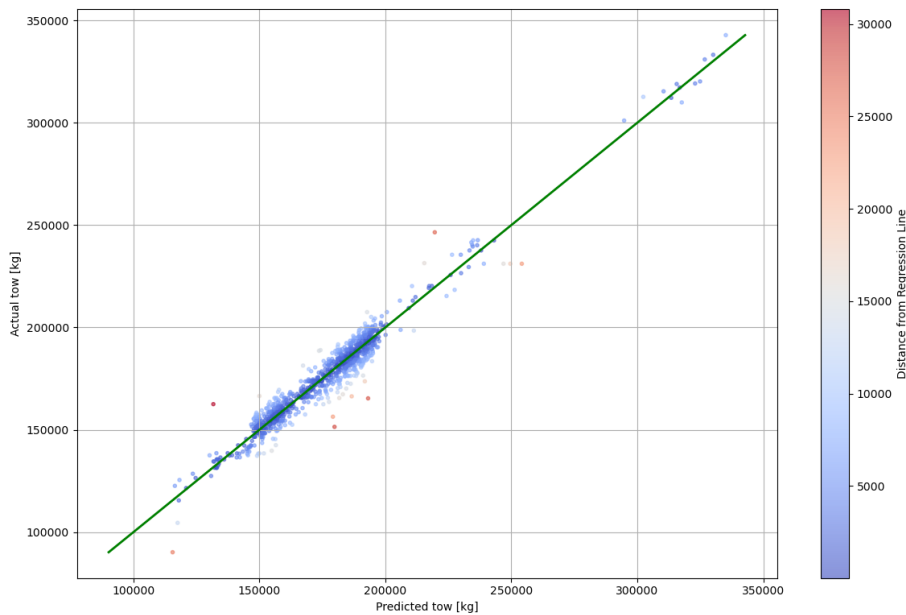


Figure 7.9: Regression Overview of Case Study: Flights with Heavy (H) Wake Category.

Figure 7.10 illustrates the distribution of the residuals which appears to be approximately normal and centred around zero. There are more atypical values with associated extreme residuals values, differently from the two other cases. Although the distribution seems somewhat narrow and peaked, this suggests the model residuals are also generally well-behaved but may have some minor deviations from perfect normality.

Figure 7.11 shows residuals against predicted values. The plot reaffirms that predictions are less reliable for higher t_{ow} values because the range of potential errors is larger. There are no linear patterns in this case how happened with the other two cases. This may indicate that either that actual tow estimations were better developed for Heavy Wake flights or that as the absence of training data these patterns didn't formed.

7.3.1 Results Analysis

The results of Table 7.1 showed a good predictive performance across all models, particularly due to the inclusion of features derived from the climb phase. XGBoost

Results

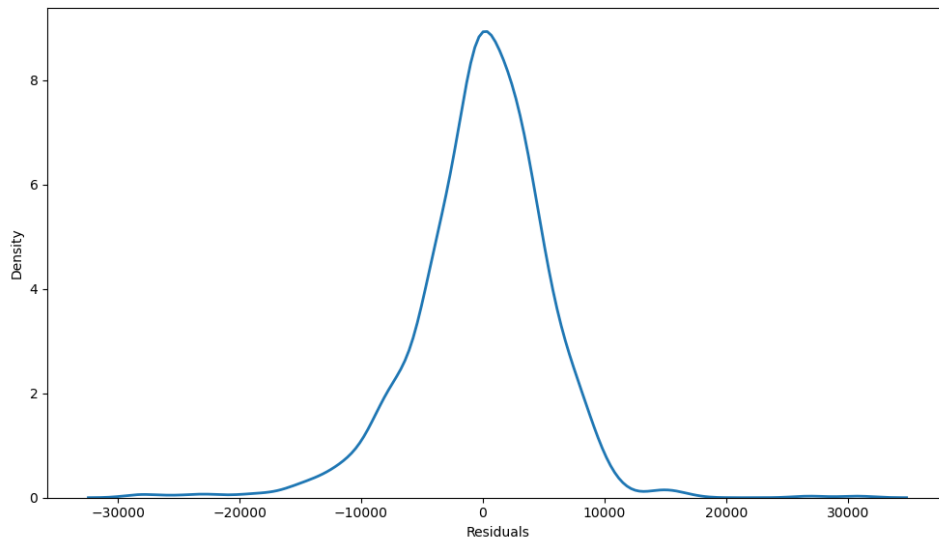


Figure 7.10: Residuals Distribution of Case Study: Flights with Heavy (H) Wake Category.

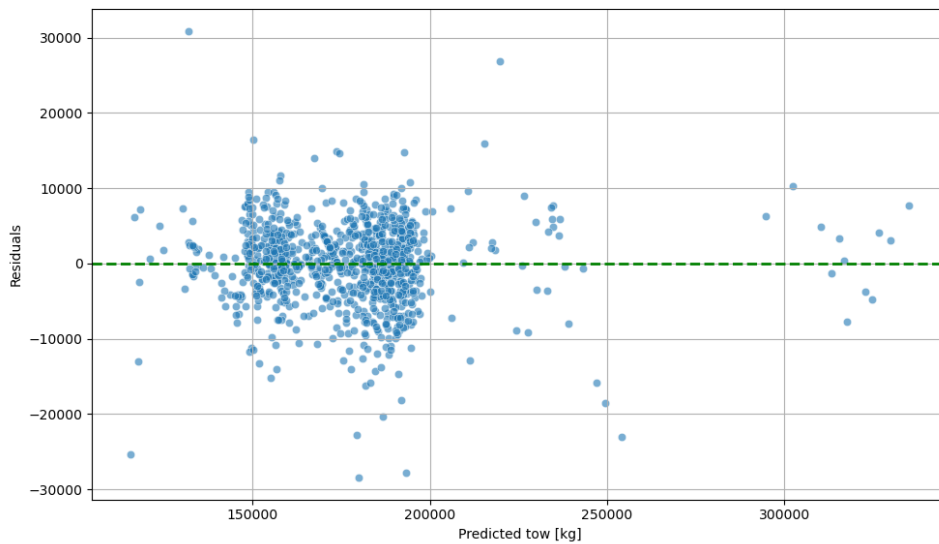


Figure 7.11: Residuals against Predictions Scatter plot of Case Study: Flights with Heavy (H) Wake Category.

stood out slightly with an RMSE of 5233.532, an MAE of 3793.143, and an R^2 of 0.956, closely followed by CatBoost and LightGBM, both with the similar MAE and MAPE values (0.021- 0.022).

Notably the gap between MAE and MAE_train is the smallest for XGBoost (2135.01), indicating a generalization gap and a significant sign of overfitting. The other two models are also likely to present overfitting, because they have a gap of 2006.55 and 1343.90 for LightGBM and CatBoost respectively.

This positions XGBoost as the least bad model for this case, for its low error values. In terms of overfitting, a clearer gap was observed for all the models. While the gaps were not completely extrem, they suggest an increase in model variance, due to

7.3. Flights with Heavy (H) Wake Category Case

fewer samples and greater heteroscedasticity among the heavy wake flights, and the absence of trajectory data.

Table 7.3: Comparison of model performance and hyperparameters for Flights with Heavy (H) Wake Category.

Algorithm	Best Hyperparameters	RMSE	MAE	MAE_train	MAPE	R ²
XGBoost	learning_rate=0.005, max_depth=7, min_child_weight=1, gamma=0.008, subsample=0.601, colsample_bytree=0.632, reg_alpha=0.434, reg_lambda= 0.062	5233.532	3793.143	1658.125	0.021	0.956
LightGBM	learning_rate=0.020, num_leaves=55, min_child_samples=21, min_child_weight=0, subsample=0.646, colsample_bytree=0.702, reg_alpha=0.190, reg_lambda=0.001	5643.614	3916.866	1910.307	0.022	0.949
CatBoost	learning_rate=0.015, depth=5, random_strength=10.234, colsample_bylevel=0.956, min_data_in_leaf=34, leaf_estimation_iterations=6	5217.831	3877.127	2533.230	0.022	0.956

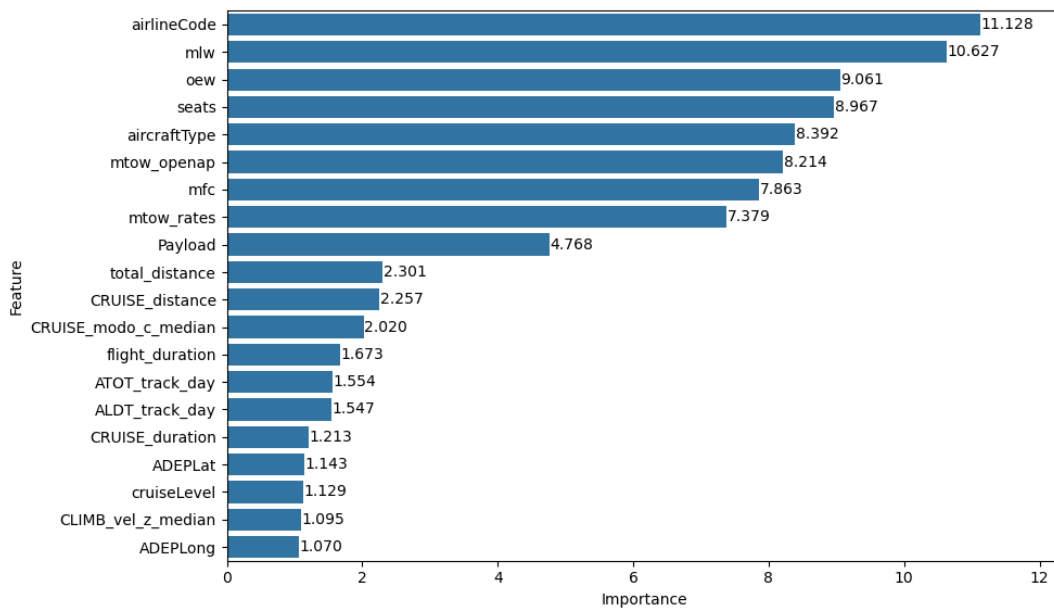


Figure 7.12: Top 20 feature importances for CatBoost of Case Study: Flights with Heavy (H) Wake Category.

7.4 Results applicability

The developed prediction models demonstrate varying levels of applicability across different operational scenarios, with their effectiveness directly linked to data availability and aircraft characteristics. The models show strongest applicability for medium wake category aircraft when climb phase trajectory data is available, achieving high predictive accuracy with RMSE values of approximately 1,300 and R^2 values exceeding 0.96. This level of performance makes these models particularly suitable for real-time operational applications where accurate t_{ow} predictions are critical for air traffic management, fuel planning, and safety assessments. For medium wake category flights without climb phase trajectory data, the models remain operationally viable despite reduced accuracy, with RMSE values increasing to approximately 1,750 while maintaining R^2 values above 0.95. This degradation in performance suggests that while climb phase data significantly enhances prediction quality, the models can still provide meaningful estimates using alternative features such as payload and cruise phase parameters. This capability is particularly valuable for historical data analysis or scenarios where real-time climb phase tracking is not available.

The applicability becomes more limited for heavy wake category aircraft, where prediction errors increase substantially with RMSE values exceeding 5,000, though R^2 values remain respectable at 0.95. The higher absolute errors, while maintaining good correlation with actual values, indicate that these models may be more suitable for trend analysis and comparative studies rather than precise parameter prediction. The increased variance in predictions for extreme t_{ow} values (both very high and very low values) suggests particular caution when applying these models to aircraft operating at the boundaries of their weight capacity. The feature importance analysis reveals that aircraft-specific characteristics consistently emerge as the most critical predictors across all scenarios, indicating strong model applicability for fleet management and aircraft-specific optimization strategies. The significance of physics-based features, particularly climb velocity and cruise altitude parameters, demonstrates the model's capability to capture fundamental aerodynamic relationships, making them suitable for applications requiring physically interpretable predictions.

The observed overfitting patterns, particularly evident in the heavy wake category case, suggest that model applicability should be carefully evaluated based on the similarity between training and deployment scenarios. The smallest generalization gaps observed in the medium wake category with climb phase data indicate this configuration offers the most robust applicability for an operational environment. For heavy wake category applications, the models may require additional validation and potentially ensemble approaches to improve reliability in production environments. These findings position the developed models as most applicable for medium-sized aircraft operations with comprehensive flight data availability, while suggesting that heavy aircraft applications may benefit from additional data sources to achieve operational-grade accuracy.

Chapter 8

Conclusions and Future Work

This thesis addressed the critical challenge of accurately estimating aircraft TOW within the ATM domain, a key factor for improving trajectory prediction, optimizing fuel consumption, and advancing environmental sustainability in aviation. The primary objective was to design, implement, and evaluate a suite of Artificial Intelligence models capable of retrospectively predicting TOW by integrating diverse data sources, including operational flight data from CRIDA, public aircraft performance specifications, and TOW estimates from the EUROCONTROL Performance Review Commission (PRC) Data Challenge.

To achieve this, a comprehensive methodology was established, beginning with an extensive data preprocessing and feature engineering phase. Data from Flight Plans (FPLs), flight trajectories, and aircraft specifications were merged and enriched with new features describing temporal, spatial, and dynamic aspects of each flight, with a particular focus on the climb phase. Exploratory Data Analysis revealed significant data imbalances, most notably a heavy skew towards Medium (M) wake category aircraft. To mitigate this, the research was structured around three distinct case studies: (1) Medium-wake flights with available climb phase trajectory data, (2) Medium-wake flights without climb trajectory data, and (3) Heavy (H) wake category flights. This partitioning allowed for the development of specialized models tailored to the unique characteristics and data availability of each scenario, utilizing state-of-the-art Gradient Boosting Decision Tree (GBDT) algorithms: XGBoost, LightGBM, and CatBoost, along with the H2O AutoML framework for automated model discovery.

The results demonstrated that the developed models can predict TOW with a high degree of accuracy, thereby successfully meeting the core objective of the thesis. For medium-wake category flights with available climb phase data, the models achieved excellent performance, with an R^2 value of approximately 0.967 and a Root Mean Squared Error (RMSE) as low as 1,333. This outcome validates the hypothesis that early flight dynamics, particularly vertical velocity during climb, are powerful predictors of aircraft mass. When climb data was absent, predictive accuracy saw a modest decline (RMSE around 1,755), but the models remained robust by leveraging other critical features like payload and cruise parameters, proving their adaptability to varying levels of data granularity. This confirms that while climb trajectory data is highly valuable, meaningful TOW estimation is still viable without it.

The analysis of Heavy-wake (H) category aircraft presented greater challenges, pri-

marily due to data scarcity and higher heteroscedasticity in flight operations. While the models maintained a strong correlation with actual TOW (R^2 around 0.956), the absolute prediction errors were substantially larger (RMSE around 5,233), and signs of overfitting were more pronounced. This finding underscores the difficulty of modelling less frequent and more varied flight profiles and highlights a key limitation of the current dataset. Nonetheless, the models for this category are still valuable for trend analysis and strategic assessments, even if not yet suitable for precise, real-time operational use. Across all cases, aircraft-specific features such as MTOW, MLW, and OEW consistently ranked as the most important predictors, reinforcing the physical and operational constraints governing aircraft weight.

In conclusion, this research demonstrates the significant potential of applying AI techniques to overcome the problem of TOW data unavailability in ATM. The resulting models provide a robust, data-driven solution that can directly support fuel consumption to achieve strategic goals outlined in ENAIRE's Flight Plan 2025, by enabling more accurate trajectory predictions. Future research should prioritize collaborative efforts to access more diverse and extensive datasets, develop robust validation methods for ML models in the absence of direct ground truth, and explore innovative hybrid approaches that leverage the strengths of physics models, fuelling and advanced ML techniques to achieve more accurate and generalizable TOW estimations. This opens up several high-impact research avenues, including modelling airline-specific fuelling strategies like tankering, creating dynamic mass profiles to enhance 4D trajectory prediction accuracy, and quantifying the environmental cost associated with carrying excess reserve fuel.

References

- [1] International Civil Aviation Organization, “ASPIRE (Aviation System Performance Improvement with Reduced Emissions) Overview and Implementation,” International Civil Aviation Organization (ICAO), Tech. Rep. IP/12 - AI 3.6, Mar. 2014, Presented at the Twenty-Fifth Meeting of the Asia/Pacific Air Navigation Planning and Implementation Regional Group (APANPIRG/25). [Online]. Available: <https://www.icao.int/APAC/Meetings/2014%20APANPIRG25/IP%2012%20-%20AI%203.6%20-%20ASPIRE.pdf> (visited on 07/03/2025).
- [2] D-ATM Partnership, “D-ATM Safety Policy v1.1,” D-ATM Partnership, Tech. Rep., Feb. 2012, Version 1.1. [Online]. Available: https://kdc-mainport.nl/wp-content/uploads/2011/04/D_SP_12_047-120223-v1-1.pdf (visited on 07/03/2025).
- [3] French Ministry for the Ecological Transition, *EGNOS – english version*, https://www.ecologie.gouv.fr/sites/default/files/documents/EGNOS_version_anglais_0.pdf, Accessed: 2025-07-03, 2025.
- [4] International Civil Aviation Organization (ICAO), *CAEP Inputs to the 2025 CORSIA Periodic Review*, <https://www.icao.int/environmental-protection/CORSIA/Documents/Forms/AllItems.aspx>, Accessed: 2025-07-03, 2023.
- [5] SESAR Joint Undertaking, *Sesar – single european sky atm research*, <https://www.sesarju.eu/>, Accessed: 2025-07-03, 2025.
- [6] Federal Aviation Administration (FAA), *Next generation air transportation system (nextgen)*, <https://www.faa.gov/nextgen>, Accessed: 2025-07-03, 2025.
- [7] ENAIRE, *Plan de vuelo 2025*, https://www.enaire.es/sobre_enaire/conoce_enaire/estrategia_empresarial/plan_de_vuelo/plan_de_vuelo_2025, Accessed: 2025-07-03, 2024.
- [8] European Commission, *Single European Sky*, https://transport.ec.europa.eu/transport-modes/air/single-european-sky_en, Accessed: 2025-07-09.
- [9] European Commission, *Mobility strategy*, https://transport.ec.europa.eu/transport-themes/mobility-strategy_en, Accessed: 2025-07-09.
- [10] EUROCONTROL, *About us*, Accessed: 2025-06-04. [Online]. Available: <https://www.eurocontrol.int/about-us>.
- [11] International Civil Aviation Organization (ICAO), *Eurocontrol*, Accessed: 2025-06-04. [Online]. Available: <https://applications.icao.int/postalhistory/eurocontrol.htm>.
- [12] ANS Performance. “PRC Data Challenge.” Accessed: 2025-06-25. (2025), [Online]. Available: <https://ansperformance.eu/study/data-challenge/> (visited on 06/25/2025).

-
- [13] M. Strohmeier, X. Olive, J. Lübke, M. Schäfer, and V. Lenders, "Crowdsourced air traffic data from the OpenSky Network 2019–2020," *Earth System Science Data*, vol. 13, no. 2, pp. 357–366, 2021. DOI: 10.5194/essd-13-357-2021. [Online]. Available: <https://essd.copernicus.org/articles/13/357/2021/>.
 - [14] SKYbrary. "Flight Plan." Accessed: 2025-06-25, SKYbrary. (2025), [Online]. Available: <https://skybrary.aero/articles/flight-plan> (visited on 06/25/2025).
 - [15] E. Arts, "Automated detection of climb phases in flight trajectories," Accessed: 2025-07-04, Master's thesis, Delft University of Technology, 2021. [Online]. Available: <https://elib.dlr.de/141738/1/EmyArtsMasterthesis.pdf>.
 - [16] U. N. Aeronautics and S. A. (NASA), *Aircraft performance models for trajectory prediction*, https://www.nasa.gov/sites/default/files/atoms/files/aircraft_performance_models_for_trajectory_prediction.pdf, Accessed: 2025-07-03, 2011.
 - [17] EUROCONTROL, *Performance review report: The importance of weight information in atm performance modelling*, <https://www.eurocontrol.int/sites/default/files/2021-10/prc-weight-information-in-atm-performance-modelling.pdf>, Accessed: 2025-07-03, 2021.
 - [18] AviationFile, *Which flight stage burns the most fuel?* Accessed: 2025-06-12, 2022. [Online]. Available: <https://www.aviationfile.com/which-flight-stage-burns-the-most-fuel/>.
 - [19] SherpaReport, *Fuel burn rates for private aircraft*, Accessed: 2025-06-12, 2023. [Online]. Available: <https://www.sherpareport.com/aircraft/fuel-burn-private-aircraft.html>.
 - [20] AviationFile. "Which flight stage burns the most fuel?" Accessed: 2025-06-12. (2022), [Online]. Available: <https://www.aviationfile.com/which-flight-stage-burns-the-most-fuel/>.
 - [21] NASA Glenn Research Center. "Thrust force." Accessed: 2025-06-12, NASA. (2023), [Online]. Available: <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/thrust-force/>.
 - [22] R. Alligier, D. Gianazza, and N. Durand, "Ground-based estimation of the aircraft mass, adaptive vs. least squares method," in *Proceedings of the 10th US-A/Europe Air Traffic Management Research and Development Seminar, ATM 2013*, Jun. 2013, pp 1–10.
 - [23] Y. S. Chati and H. Balakrishnan, "Statistical Modeling of Aircraft Takeoff Weight," in *Twelfth USA/Europe Air Traffic Management Research and Development Seminar (ATM 2017)*, Accessed: 2025-07-09, Seattle, Washington, Jun. 2017. [Online]. Available: <https://dspace.mit.edu/handle/1721.1/115247>.
 - [24] G. Vourros, T. Tranos, K. Blekas, G. Santipantakis, M. Melgosa Farrés, and X. Prats Menéndez, "Data-driven estimation of flights' hidden parameters," in *12th SESAR Innovation Days: Inspiring long-term research in the field of air traffic management*, Budapest, Hungary: Single European Sky ATM Research (SESAR), Dec. 2022, pp. 1–6. [Online]. Available: <http://hdl.handle.net/2117/386838>.
 - [25] R. Alligier and D. Gianazza, "Learning aircraft operational factors to improve aircraft climb prediction: A large scale multi-airport study," *Transportation Research Part C: Emerging Technologies*, vol. 96, pp. 72–95, 2018, ISSN: 0968-090X. DOI: <https://doi.org/10.1016/j.trc.2018.08.012>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0968090X18311896>.

REFERENCES

- [26] X. He, F. He, X. Zhu, and L. Li, "Data-driven method for estimating aircraft mass from quick access recorder using aircraft dynamics and multilayer perceptron neural network," *ArXiv*, vol. abs/2012.05907, 2020. [Online]. Available: <https://api.semanticscholar.org/CorpusID:228376122>.
- [27] J. Sun, J. Ellerbroek, and J. Hoekstra, "Modeling and inferring aircraft takeoff mass from runway ads-b data," Jun. 2016.
- [28] J. Sun, H. A. P. Blom, J. Ellerbroek, and J. M. Hoekstra, "Particle filter for aircraft mass estimation and uncertainty modeling," *Transportation Research Part C: Emerging Technologies*, 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:197454207>.
- [29] A. Software, *Machine Learning and Deep Learning*, <https://apro-software.com/machine-learning-and-deep-learning/>, Accessed: July 17, 2025, 2020.
- [30] S. Boutnaru, "The Artificial Intelligence Journey — Stages of Artificial Intelligence," *Medium*, 2020. [Online]. Available: <https://medium.com/@shlomiboutnaru/the-artificial-intelligence-journey-stages-of-artificial-intelligence-fc00f279564f>.
- [31] Allied Universal, *What's AI Anyway?* <https://www.aus.com/blog/whats-ai-anyway>, Accessed: July 17, 2025, 2025.
- [32] Simplilearn, *Top 10 Deep Learning Algorithms You Should Know in 2025*, https://www.simplilearn.com/tutorials/deep-learning-tutorial/deep-learning-algorithm#what_is_deep_learning, Accessed: July 17, 2025, 2025.
- [33] DeepLearning.AI and Stanford Online, *Machine Learning Specialization*, <https://www.coursera.org/specializations/machine-learning-introduction>, Offered by Coursera; Accessed: July 17, 2025, 2022.
- [34] Gaurav, "An Introduction to Gradient Boosting Decision Trees," *Machine Learning Plus*, Jun. 2021. [Online]. Available: <https://www.machinelearningplus.com/machine-learning/an-introduction-to-gradient-boosting-decision-trees/> (visited on 06/16/2025).
- [35] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*, 3rd. Sebastopol, CA: O'Reilly Media, 2022, ISBN: 978-1098125974.
- [36] H. Tao, S. Salih, A. Oudah, *et al.*, "Development of new computational machine learning models for longitudinal dispersion coefficient determination: Case study of natural streams, united states," *Environmental Science and Pollution Research*, vol. 29, May 2022. DOI: 10.1007/s11356-022-18554-y.
- [37] A. Pal, "Gradient Boosting Trees for Classification: A Beginner's Guide," *Medium*, Mar. 2021. [Online]. Available: <https://medium.com/swlh/gradient-boosting-trees-for-classification-a-beginners-guide-596b594a14ea> (visited on 06/17/2025).
- [38] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on tabular data?" *arXiv preprint arXiv:2207.08815*, 2022. DOI: 10.48550/ARXIV.2207.08815. [Online]. Available: <https://arxiv.org/abs/2207.08815>.
- [39] M. C. R. Murça, M. R. O. A. Maximo, J. P. A. Dantas, *et al.*, "Open machine learning models for actual takeoff weight prediction," *Journal of Open Aviation Science (Pre-print)*, 2022, Article uploaded by user.

- [40] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD ’16, San Francisco, CA, USA: Association for Computing Machinery, 2016, pp. 785–794. DOI: 10.1145/2939672.2939785. [Online]. Available: <https://dl.acm.org/doi/10.1145/2939672.2939785>.
- [41] Microsoft Corporation and LightGBM Contributors, *LightGBM Documentation*, version stable, Accessed: July 17, 2025, Microsoft Corporation, 2025. [Online]. Available: <https://lightgbm.readthedocs.io/en/stable/>.
- [42] T. Keldenich, “LightGBM: A Guide,” *Inside Machine Learning*, Jan. 2023. [Online]. Available: <https://inside-machinelearning.com/en/lightgbm-guide/> (visited on 06/18/2025).
- [43] M. Joseph, “The Gradient Boosters IV: LightGBM,” *Deep & Shallow*, Feb. 2020. [Online]. Available: <https://deep-and-shallow.com/2020/02/21/the-gradient-boosters-iii-lightgbm/> (visited on 06/18/2025).
- [44] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: unbiased boosting with categorical features,” *arXiv preprint arXiv:1706.09516*, 2017. [Online]. Available: <https://arxiv.org/pdf/1706.09516>.
- [45] PlainEnglish.io, *Smart Aspects of CatBoost Algorithm*, <https://python.plainenglish.io/smart-aspects-of-catboost-algorithm-2720a6de4da6>, Accessed: July 17, 2025, Mar. 2025.
- [46] T. Akiba, S. Sano, S. Suzuki, *et al.*, “Optuna: A Next-generation Hyperparameter Optimization Framework,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, ser. KDD ’19, Anchorage, AK, USA: Association for Computing Machinery, 2019, pp. 2623–2631. DOI: 10.1145/3292500.3330701. [Online]. Available: <https://dl.acm.org/doi/10.1145/3292500.3330701>.
- [47] R. Singh, “Optuna: A Flexible, Efficient, and Scalable Hyperparameter Optimization Framework,” *Towards Data Science*, Oct. 2020. [Online]. Available: <https://towardsdatascience.com/optuna-a-flexible-efficient-and-scalable-hyperparameter-optimization-framework-d26bc7a23fff/> (visited on 06/19/2025).
- [48] H2O.ai, *H2O AutoML Documentation*, version latest-stable, Accessed: July 17, 2025, H2O.ai, 2024. [Online]. Available: <https://docs.h2o.ai/h2o/latest-stable/h2o-docs/automl.html>.
- [49] H2O.ai, *Data Science Documentation*, version latest-stable, Accessed: July 17, 2025, H2O.ai, 2024. [Online]. Available: <https://docs.h2o.ai/h2o/latest-stable/h2o-docs/data-science.html>.
- [50] Databricks, *Mlflow on databricks*, <https://docs.databricks.com/aws/en/mlflow/>, Accessed: 20 June 2025, Databricks.
- [51] OpenAP. “OpenAP.dev.” Accessed: 2025-06-25. (2024), [Online]. Available: <https://openap.dev/> (visited on 06/25/2025).
- [52] pandas development team, *pandas.merge_asof — pandas 2.3.0 documentation*, https://pandas.pydata.org/docs/reference/api/pandas.merge_asof.html, Accessed: 2025-07-02, 2025.
- [53] Airbus, *A220-100 | A220 | Aircraft | Airbus*, <https://aircraft.airbus.com/en/aircraft/a220/a220-100>, Accessed: 2025-07-02, 2025.
- [54] Airbus, *A220-300 | A220 | Aircraft | Airbus*, <https://aircraft.airbus.com/en/aircraft/a220/a220-300>, Accessed: 2025-07-02, 2025.

REFERENCES

- [55] Aircraft Commerce. "Flight operations it systems." Accessed: 2025-07-02. (2022), [Online]. Available: https://www.aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs/General%20Articles/2022/142_FLTOPS_A.pdf.
- [56] Snowflake Inc., *HAVERSINE function — Snowflake Documentation*, [Online; accessed 13-July-2025], 2025. [Online]. Available: <https://docs.snowflake.com/en/sql-reference/functions/haversine>.
- [57] R. Al-Rfou, J. Al-Azzeh, and A. Al-Tarawneh, "Exploring the impact of financial crises on the stock market using machine learning techniques," *Journal of Big Data*, vol. 7, no. 1, pp. 1–17, 2020. DOI: 10.1186/s40537-020-00369-8.